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5G RF Antenna optimization Beamforming and MIMO

June 2025

Agenda

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- Beamforming Concepts
- Beamforming Management
- Performance Management
- General view on Antenna Optimization
- Beamforming Introduction
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Introduction- Problem Statement

Introduction

Motivation

The RF optimization is a foundation for a correct working of the network. It should allow the right level of performances that will make benefit the end-user experience.
The main areas of activity related to the RF optimization are related to:

- Coverage footprint
- Signal quality
- General interference management

5G technology benefits of additional techniques like the beamforming: the beamset configuration is a little bit beyond the traditional RF optimization of the transmitted power, antenna tilt and azimuth, but it is very important because its parametrization impacts the antenna characteristics and is integral part of the 5G RF optimization activities.

Introduction

Check of 5G radio coverage

The 5G radio coverage analysis is an activity that is often performed during the early stages of a network deployment, but it is structurally part of the optimization after the network acceptance;
Some areas could require a check of the received power for different reasons like change of the traffic distribution, new distribution of the final users, deployment of new sites, etc. and in general after any change of the antenna configuration.

- In general NSA configuration requires the overlap between the 5G on the LTE coverage to enable the control plane connectivity through the 4G layer.

- The most typical measurement considered for the coverage evaluation is the SS-RSRP that is the measured level of the received power on the SS+PSS+PBCH part of the SSB.

Introduction

Beamforming feature and its impact on improving Radio conditions

- Beamforming functionality is embedded in the **5G specifications** from the beginning.

- The immediate advantage is that the **coverage is extended** because the transmitted signal can be better directed towards the receiving antenna. A beneficial side effect is that the **interference in the network is reduced** because the transmitted signal is not directed towards locations where it is not needed.

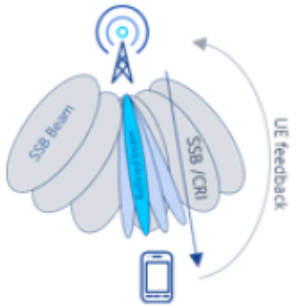
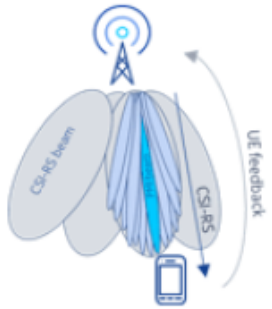
- The achievable coverage gains increase with the number of antenna elements, and the size of the antenna elements is naturally dependent on the used frequency band. In practice this means that frequency **bands below 1 GHz do not use beamforming** (except for the traditional static vertical beam forming which is generally used also by 2G, 3G and 4G BTSs), that other frequency bands in the **FR1 frequency range generally use beamforming (digital)** and that it is **pretty much mandatory for mmWave** frequency bands to use beamforming (analog beamforming).

- The interference reduction leads to an indirect capacity increase because less interference means that higher coding schemes can be used. In future releases, the combination of **beamforming and multi-user MIMO features will also give a direct capacity gain** because it will be possible to assign same frequency and time domain resources to multiple UEs, provided they are using different beams



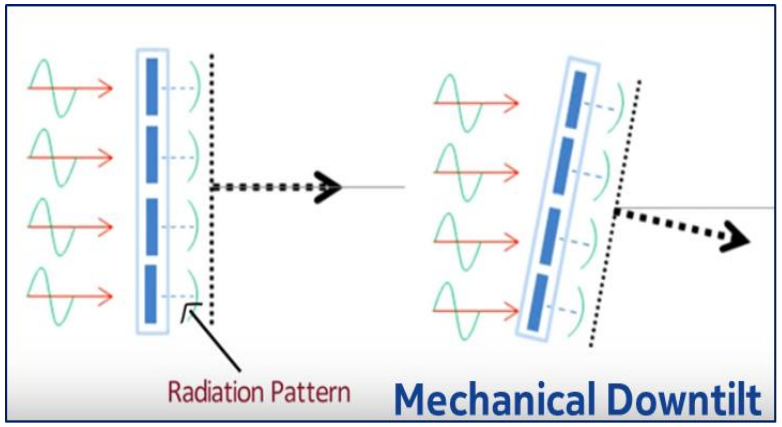
Beamforming Concepts

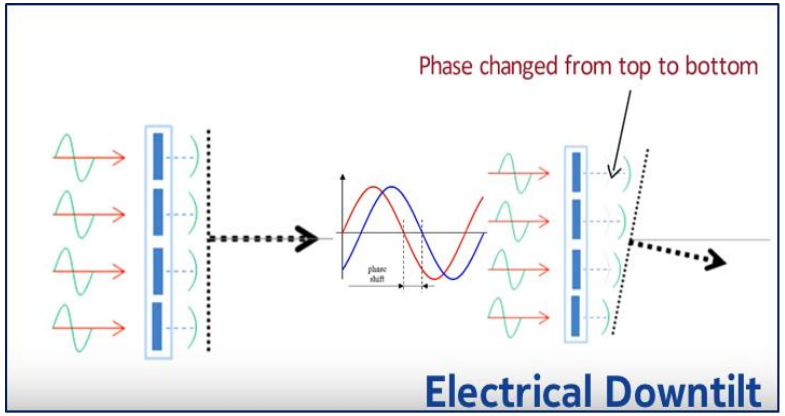
Beamforming Characteristics

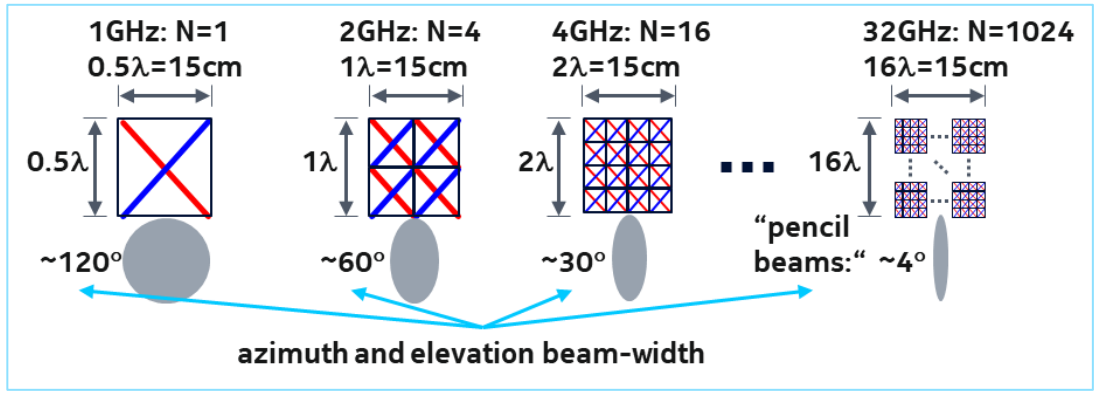
	GoB SSB BF + Refined beams	4x8 CSI-RS 3GPP Codebook BF	SRS based BF	Eigen beamforming
				
gNB antenna type	8 TRX, 32TRX, 64TRX	8 TRX, 32TRX, 64TRX	32TRX, 64TRX	32TRX, 64TRX
Cell range	>500m	>500m	200 – 500m (limited UE Tx power)	<200m (limited UE TX power + wide band SRS)
UE capability needed	Standard	+ 4 port CSI-RS	+ TAS (optional)	+ TAS (mandatory)
Number of UEs served	Medium (due UE specific CSI-RS)	High	Medium (will be used in parallel with CSI-RS & GoB)	Very Limited (will be used in parallel with CSI-RS & GoB)
User throughput				(for best radio conditions and multipath propagation)
Spectral efficiency				
Carrier aggregation	Primary and secondary cell	Primary and secondary cell	Only if primary cell *	Only if primary cell

Active Antenna Evolution & Impact of high frequency on Dipole size

More the antenna elements => sharper the beam shape => concentrated energy => Less Interfered => Great Performance

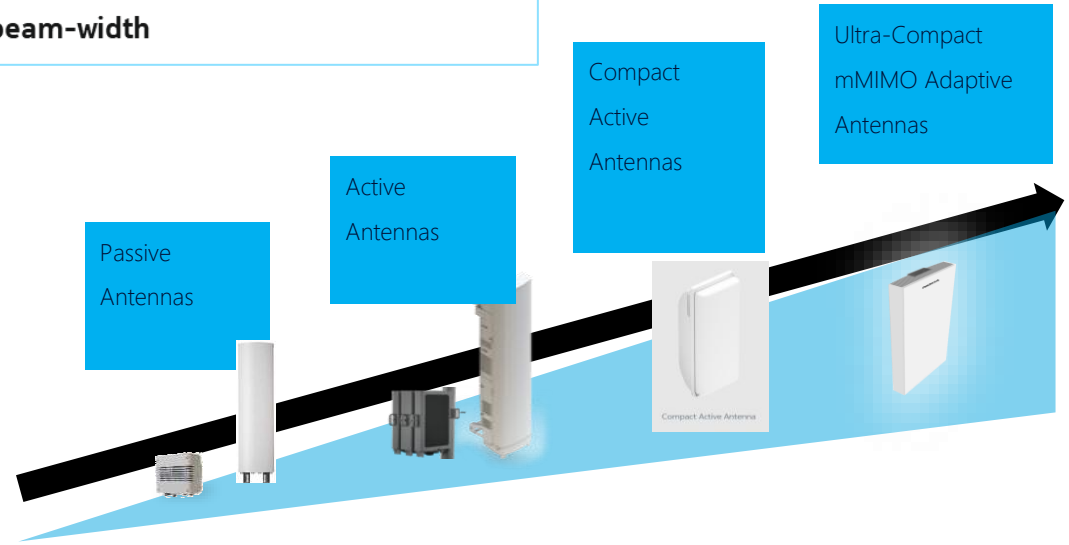




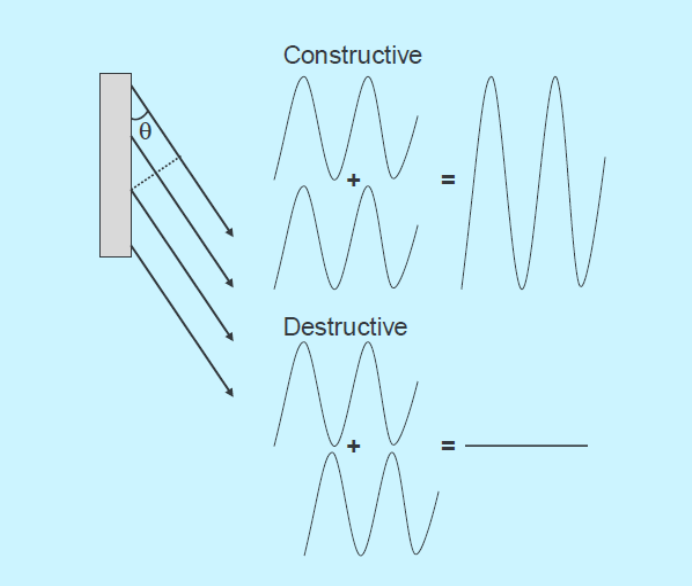


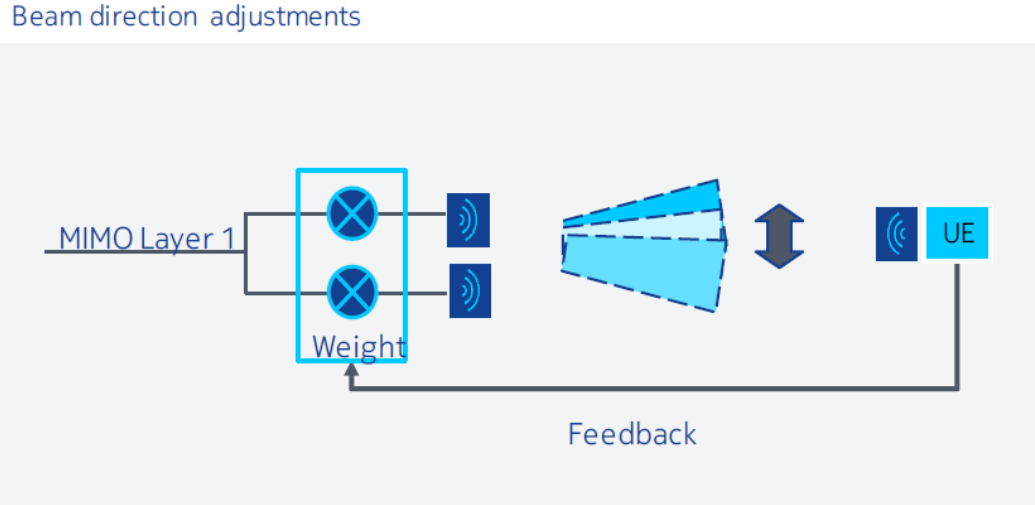
$C = f^* \lambda \rightarrow f = 1 / \lambda$

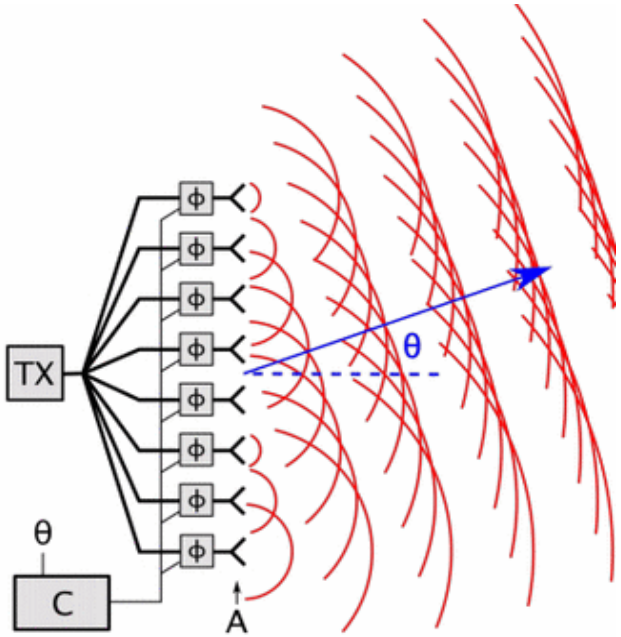
Length of a Dipole:
For effective radiation, the length of the dipole is always λ / 2



Adaptive Antenna and the governing principles







Beam direction can be adjusted by the relative phasing of the transmitters in the massive MIMO antenna.
When the multiple transmissions are received constructively by the device, the signal is amplified.
When the multiple transmissions are received destructively, the signal is zeroed out.

Beamforming management

Basic Beam Sets

Definition of basic sets of SSB

Beam sets and basic beam sets

The basic beam sets is defined using the nomenclature #k#l#m …

the number of columns in a row is given by the respective integer, preceded by the character '#'

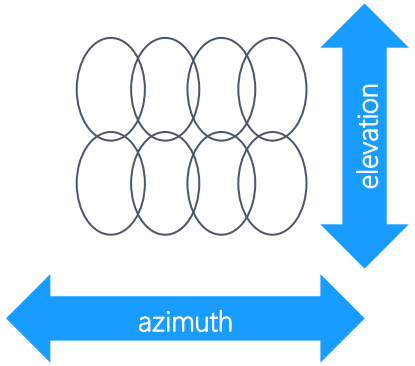
rows are counted from top to bottom, i.e., the first row is the one with highest pointing beams, the last row is the one with lowest pointing beams

Examples:

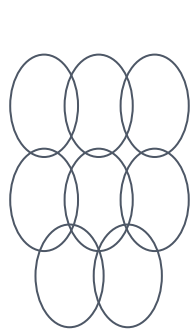
- basic beam set #4#4 denotes a beam set with 2 rows and 4 columns in each row
- basic beam set #3#3#2 denotes a beam set with 3 rows, most upper row and middle row with 3 beams each, lowest row with 2 beams only

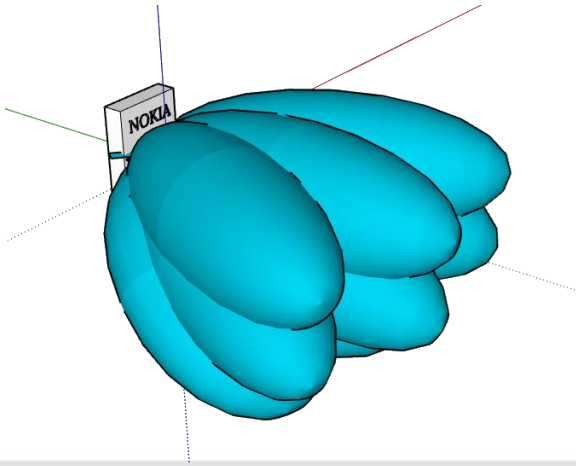
Number of beams in basic beam set **must match** with number of transmitted SS blocks set by **NRCELLGRP-numberOfTransmittedSsBlocks** parameter

basic beam set #4#4



basic beam set #3#3#2





Basic Beam Sets

Definition of basic sets of SSB

Following basic beam sets are supported by 5GC000533:

- beamSet #1
- beamSet #2
- beamSet #4
- beamSet #6
- beamSet #8
- beamSet #4#4
- beamSet#5#3
- beamSet #6#2
- beamSet #3#3#2
- beamSet #2#2#2#2

5GC001695:Extension of GoB pattern sub 6GHz supports more GoB patterns. #3#3, #5#1, #4#2, #2#2, #3#1

- Basic beam set is selected by the operator with **NRCELL-beamSet.basicBeamSet** parameter
- Selection of 90° or 120° azimuth opening angles is based on the left and right edge angles which are defined by the operator

	Beam pattern	Beam set	# SSB beams	Azimuth opening angles
1 row	0	#1	1	{45°, -45°}, {60°, -60°}
	0-0	#2	2	{45°, -45°}, {60°, -60°}
	0-0-0-0	#4	4	{45°, -45°}, {60°, -60°}
	0-0-0-0-0-0	#6	6	{45°, -45°}, {60°, -60°}
	0-0-0-0-0-0-0-0	#8	8	{45°, -45°}, {60°, -60°}
2 rows	0-0-0-0 0-0-0-0	#4 #4	8	{45°, -45°}, {60°, -60°}
	0-0-0-0-0 0-0-0	#5 #3	8	{45°, -45°}, {60°, -60°}
	0-0-0-0-0-0 0-0	#6 #2	8	{45°, -45°}, {60°, -60°}
3 rows	0-0-0 0-0-0 0-0	#3 #3 #2	8	{45°, -45°}, {60°, -60°}
4 rows	0-0 0-0 0-0 0-0	#2 #2 #2 #2	8	{20°, -20°}, {45°, -45°},

Basic Beam Sets

Beam Refinement

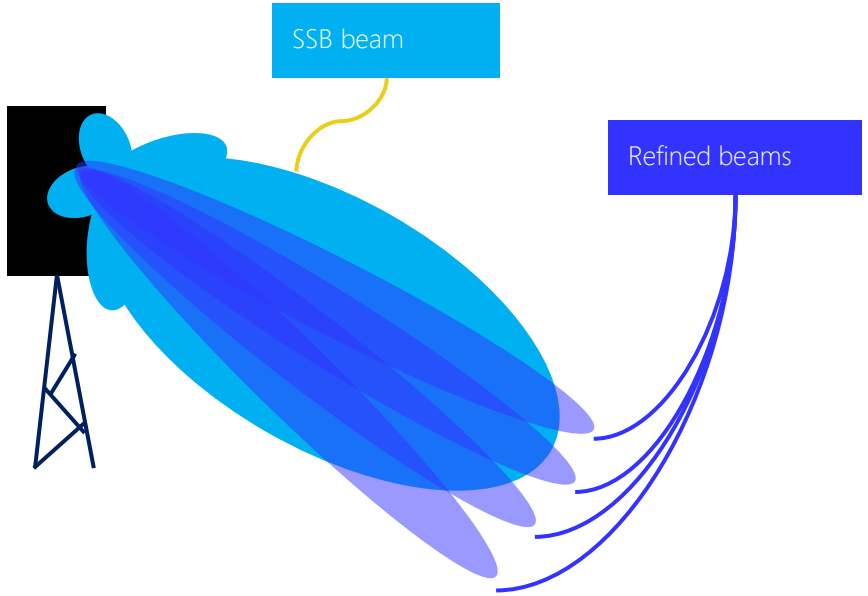
In case of operating with a grid of beam (GoB) and especially with carrier frequencies below 6GHz with limited number of SSB beams, beam refinement on 5G-NB side is used to achieve higher SINR for a single UE in both DL and UL direction, as well as better separation of UEs in case multiple UEs are served on the same time and frequency resources, i.e., MU-MIMO operation.

To enable execution of respective measurements by UEs, CSI-RS have to be transmitted for beam refinement. The antenna ports used for these CSI-RS have to be mapped to the beams used for refining a respective SSB beam

4 refined beams per SSB beam are implemented.

From 5G19B, gNB will check UE's refine beam capability (***beamManagementSSB-CSI-RS → maxNumberCSI-RS-Resource)***, in case of beamRefinementP2=true, if
(UE's maxNumberCSI-RS-Resource less than numberOfRefinedBeams) then
beamRefinementP2 fall back to value 0 and applied BeamRefinementP2=false to this UE.

Beam refinement is activated by setting parameter NRCELL-beamSet.nrBtsBeamRefinementP2 = true



Basic Beam Sets

Beam Refinement

Basic beam refinement patterns in digital beamforming

For beam refinement, three different patterns are foreseen:

Square refinement

Square refinement is useful especially for SSB beams which have a mostly circular slope

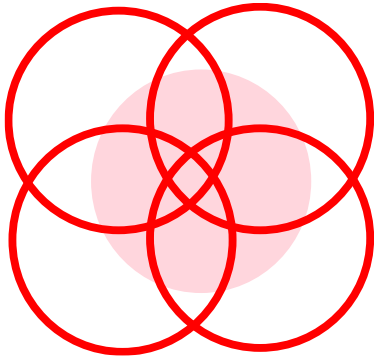
Azimuth refinement

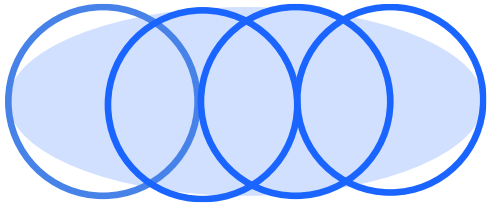
Azimuth refinement is appropriate mainly for elliptic beams with larger axis in azimuth direction.

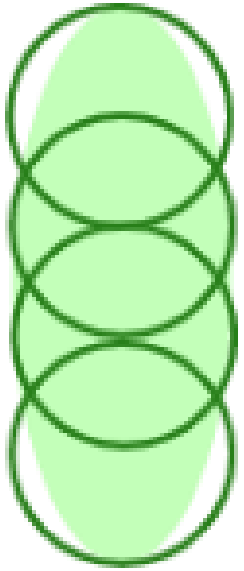
Elevation refinement

Elevation refinement is appropriate mainly for elliptic beams with larger axis in elevation direction.

Note: Elevation refinement is not supported in the basic beam sets

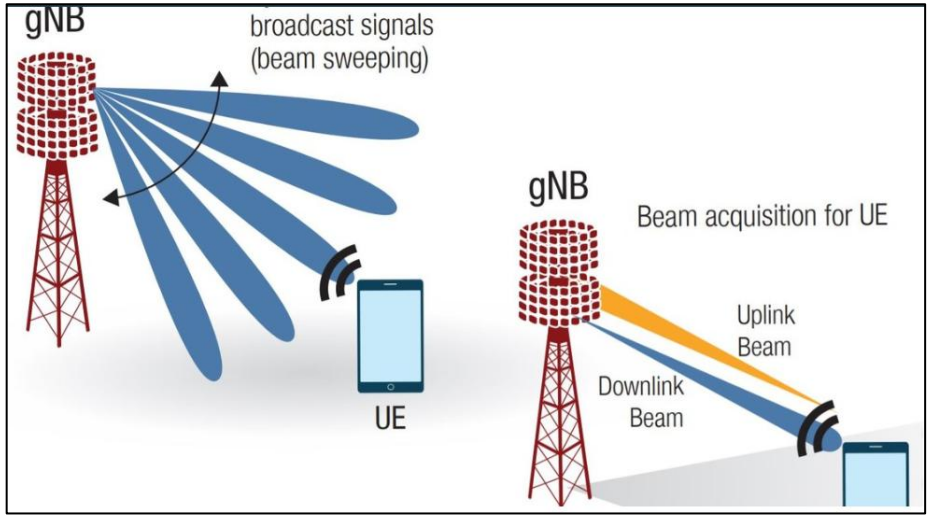


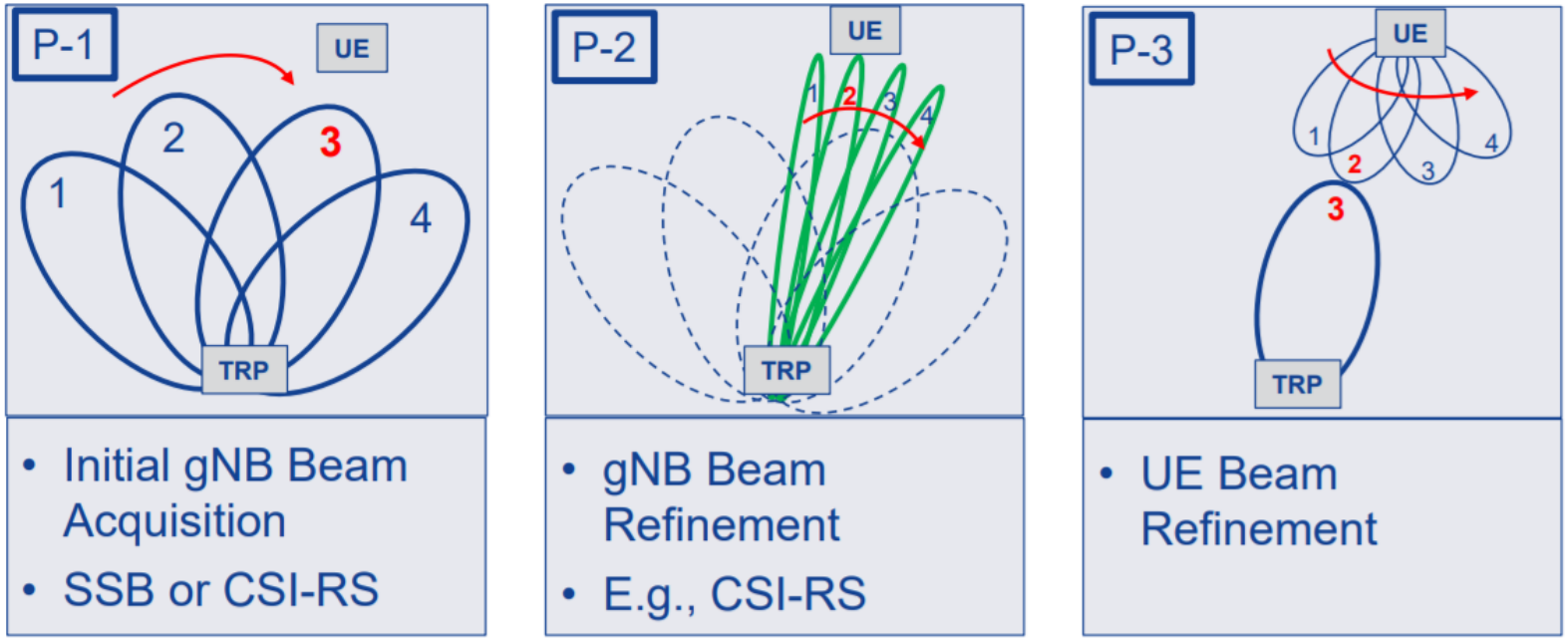




Beam Management

- Downlink beam management includes 3 general steps for beam selection
- These are known as P1, P2 and P3
- P2 is most applicable to digital beamforming which has relatively wide beams for the initial acquisition





Beamforming Features & Recommendations

Feature Name	Starting Release	Recommendations	Beam type configuration
5GC000533 Digital Beamforming for CPRI-based Rus	5G19/SRAN 21B	Activate to have the Beam sets and Basic beam sets up to 8 SSB beams. This feature allows the activation of the Beam refinement type as well.	SSB beams / Refined beams
5GC000534 Digital Beamforming for eCPRI-based Rus	5G20A/SRAN 21B	Activate to have the Beam sets and Basic beam sets up to 8 SSB beams. This feature allows the activation of the Beam refinement type as well.	SSB beams / Refined beams
5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO	5G20B/SRAN 21B	Activate to have up to 4 beamformed CSI-RS resources (CSIRS beams) where each CSI-RS resource conveys 8 ports. This feature allows the UE to choose a PMI beam using the 8-port codebook and up to 256 PMI beams.	SSB beams / CSI-RS beams / PMI beams
5GC001866 Delivery of beam set realization to gNB	5G20B	Just in case, is required to deliver new beam set definitions to gNB, without the need to upgrade the system release.	SSB beams / Refined beams
5GC000524 Downlink MU-MIMO for Digital Beamforming	5G21A/SRAN 21B	5G21A introduced DL MU-MIMO with 5GC000524 but the feature was not commercialized due to the limitations. All drawbacks of the feature 5GC000524 are resolved in the CB008305 framework.	SSB beams / CSI-RS beams / PMI beams
5GC001086 SRS based beamforming in UL in the network	22R2-SR	Out of the scope at the time from 9955 beams set optimization procedure, due to not available the DFT's beams with the current tools (e.g., 9955 PMI beam optimization).	SRS beams (2D DFT beamsets)
CB008247 SRS based downlink beamforming	22R3-SR	Out of the scope at the time from 9955 beams set optimization procedure, due to not available the DFT's beams with the current tools (e.g., 9955 PMI beam optimization).	SRS beams (2D DFT beamsets)
CB008305 Downlink MU-MIMO enhancement	22R4-SR	CB008305 DL MU-MIMO Enhancement cannot coexist with legacy 5GC000524 DL MU-MIMO for the Digital BF feature. CB008305 DL MU-MIMO Enhancement is based on CB0008247 DL SRS-based BF feature.	SRS beams (2D DFT beamsets)
CNI-84878 Beam power balancing optimizations for CB008247 and adaptions for CB008305	22R4-SR	CB008247 algorithms optimization improving DL performance of CB008247 and CB008305 features. This CNI will cover additional CB008247 performance optimizations that did not fit the 22R3 time schedule. New algorithms will be adapted to work with DL MU-MIMO (CB008305) but in the current implementation (22R4) CNI does not impact CB008305 operation regardless CNI is enabled or not.	SRS beams (2D DFT beamsets)
CB010602 SRS based Beamforming extension	23R1-SR	Out of the scope at the time from 9955 beams set optimization procedure, due to not available the DFT's beams with the current tools (e.g., 9955 PMI beam optimization). CB0010602 replaces CNI-84878 which partially introduced beam power balancing in 22R4.	SRS beams (2D DFT beamsets)
CB010388 Support of 5GC001086 and CB008247 using 4DL4UL cell type and 64TRX Rus	23R2-SR	Out of the scope at the time from 9955 beams set optimization procedure.	SRS beams (2D DFT beamsets)

Performance Management

Beam Management

Some **KPIs at map level** can be considered during the antenna optimization tasks like:

- SS-RSRP
- SS-RSRQ
- Best Serving Beams (distribution)

These KPI maps make able to evaluate the distribution of coverage, quality and beams usage in a geographycal area

Additional information coming from OSS can be used to understand the performances, example of groups of counters to be considered are:

Active cell Throughput at MAC level (DL/UL) (KPI NR5088-5090)

- **low levels** can show if either low MCS or low MIMO Rank are used

- **DL/UL quality signal counters (**MCS and RANK belonging to the group 55301 and related KPIs) show the quality of the downlink signal. Through thes counters it is possible to check if the implemented antenna configuration changes are improving or degrading the cell performances
- Specific beamforming counters and KPIs are defined in the series 55305

Beam Management

PM Counters

- DL Serving Beam ID Histogram
- UL Serving Beam ID Histogram
- Serving Beam ID before HO Histogram
- Beam Change Histogram
- Failed Beam Change Histogram
- Beam Change Target Histogram
- Best and Second Best Beam Histogram
- UE reported differential L1-RSRP of second best beam
- Beam Recoveries
- Succesfull Beam Recoveries
- Time of "UE on single beam" histogram
- Number of UEs per beam
- Number of beam toggles

Beamforming counters

Beamforming counters are available in two measurements:

- M55305 NR Beamforming (SSB, PDSCH and PUSCH)
- M55306 NR Rach Usage. Counters for contention-based and for dedicated preambles

They can be used to

- Calculate angular distribution of the UEs in the cell
- Calculate the mobility of UEs in the cell (via the number of beam switches)
- Understand if certain areas of the cell have bad radio coverage (via the beam change failure and the beam recovery counters)
- Triangulate to locate areas of (PRACH) interference (via the M55036 counters)

There are still certain limitations in the possible analysis

- Timing advance counters are not available per beam id, so UE location cannot be determined
- Signal quality counters (SINR, RSRP, CQI) are not available per beam id, so the quality of a specific beam can only be indirectly determined via the beam change failure and beam recovery counters

General performance analysis methods

Drive tests

The **best and most expensive** method **to find RF problems** in the network is to make traditional drive tests.
The methodology is the same as used for pre-launch optimization, and is described in the pre-launch optimization engineering guideline.

Apart from being expensive, **there are other major drawbacks** of this method:

- It is straightforward to measure street coverage but **measuring indoor coverage** (where the majority of the subscribers can be expected to be) can be a challenge. Street coverage can be very difficult to map to real user coverage, especially in a city environment with many high-rise buildings.

- Drive tests give a snapshot of the coverage and interference situation. Especially the interference will vary dynamically with the traffic load and the location of the users, and the ever-ongoing changes in the network, such as additions of new sites, will also **quickly make drive test results outdated.**

- Measuring the downlink signal as is done in drive tests will not necessarily reveal problems with the **uplink signal**. If uplink signal issues are expected, it may be necessary to capture BTS traces simultaneously with the drive test.

General performance analysis methods

Drive test alternatives and complements

Predictions by network planning tool

An alternative is to base the analysis on predictions made by a radio network planning tool, such as the Atoll / 9955. The 5G Outdoor RF Design Guideline describes how to use the 9955 tool and in general how predicted performance can be calculated. Advantages of this approach is that it is a "desktop exercise" (so it is cheap) and indoor coverage, even in high-rise buildings, can be estimated. The disadvantage is that outdated maps or wrong assumptions about building penetration losses will give wrong predictions. Naturally, mistakes in the site installations, such as wrong antenna azimuths, will not be detected by this method.

Network traces

There is some possibility to use network traces for 5G RF optimization. Various PCMD features are available for NSA deployments, and it is possible to see the CQI, rank and BLER for each connection. It is also possible to analyse RSRP and RSRQ from neighbour cells (not serving cell) immediately before a handover happens. More functionality will be added in SRAN21A (LTE4776: PCMD for LTE-NR Dual Connectivity) and 5G21x / 5G22x. 5G PCMD has so far only been deployed in a few networks, so there is not yet much practical experience with using these traces. Another tool available for internal usage for analysis and optimization is MUSA.

OSS counters

In some cases, it is possible to directly determine from OSS counters what the problem is and how it can be mitigated, and in other cases OSS counters can be used as a "pointer" to a geographical area which needs to be analysed further with field tests.

General view on Antenna Optimization

Considerations for antenna and Beam Set pattern optimization

- The maximum number of coarse beams (SSB) is dependant from the TDD frame structure (could be 6 or 8 SSBs)
- Beamset configuration depends on target coverage area and traffic distribution
  - cell range, antenna height, specific buildings/areas, hotspot locations
- overheads generated by SS/PBCH and Tracking Reference Signal (TRS) (see following slides for feature 5GC002534)
- antenna gain of beams within a Beam Set
  - high gains can benefit cell edge coverage and average cell throughput
- horizontal span of beams and nulls between beams and sectors
- vertical sidelobes which become more significant as the number of rows of beams increases
- MIMO performance in terms of the UE reporting ranks 3 and 4
  - requires simultaneous reception from multiple beams
- utilisation of each beam
  - the overhead associated with a beam may not justified if it has low utilisation
- scope for Multi-User MIMO in the future
  - increased number of beams may increase number of simultaneously served UE

Example - #6#2 Beam Set

- Example of the #6#2 Beam Set from a European city



- 64T64R Active Antenna
- 2 degrees of mechanical downtilt
- tdLte Frame Structure with 3 ms time shift
- 80 MHz channel with 256QAM and 4x4 MIMO



Beam Utilisation

- Majority of coverage is provided by the upper 6 beams
- Note that the complete coverage area extends beyond the rectangular drive route region
- Even distribution of traffic between beams will be beneficial when Multi-User MIMO is supported



Antenna Optimization

The RF optimization includes possible changes of pure physical antenna aspects and configuration parameters

Antenna and parameters changes can include:

(Typical parameters considered during RF Optimization)
- Electrical tilt
- Beam pattern
- Downlink power control

(Quite seldom adopted changes)
- Azimuth angle
- Mechanical tilt

Beamset configuration

Impact

The selection of the beamset configuration as already mentioned is depending by a variety of variables (RU type, TDD frame, coverage and user distribution, intersite distance, etc.)

The beamset configuration can be selected using:

- An high level recommendation
- Using OSS bemformig KPIs
- Process based on planning tool considerations (for this process refer to the 5G OUTDOOR RF DESIGN GUIDELINES)
- Using measurements (DT or MDT) analysis with optimization tools maximizing coverage/traffic

Example of High Level Beam Patterns selection (general recommendations)

	Beam sets recommended for deployment scenario	Release	Beam sets recommended for cell size	#SSB	Basic SSB Beam Set		Recommended TDD sub-6Ghz radios*
1 row	Rural – Urban area	19A	small	1	O	#1	All 8TRX**, 32,64TRX
			small-medium	2	o-o	#2	
			medium	4	o-o-o-o	#4	
			medium-large	6	o-o-o-o-o-o	#6	
			large	8	o-o-o-o-o-o-o-o	#8	
2 rows	Urban – Dense urban area	19B	small	4	o-o o-o	#2#2	64TRX
		19B	small-medium	4	o-o-o-o o	#3#1	64TRX
		19B	small-medium	6	o-o-o o-o-o	#3#3	64TRX
		19B	small-medium	6	o-o-o-o o-o	#4#2	64TRX
		19A	small-medium	8	o-o-o-o o-o-o-o	#4#4	64TRX
		19B	medium	6	o-o-o-o-o o	#5#1	64TRX
		19A	medium	8	o-o-o-o-o o-o-o	#5#3	64TRX
		19A	medium-large	8	o-o-o-o-o-o o-o	#6#2	64TRX
3 rows	Dense urban area – High rise buildings	19A	small-medium	8	o-o-o o-o-o o-o	#3#3#2	Only 64TRX
4 rows	High-rise buildings	19A	small-medium	8	o-o o-o o-o o-o	#2#2#2#2	Only 64TRX

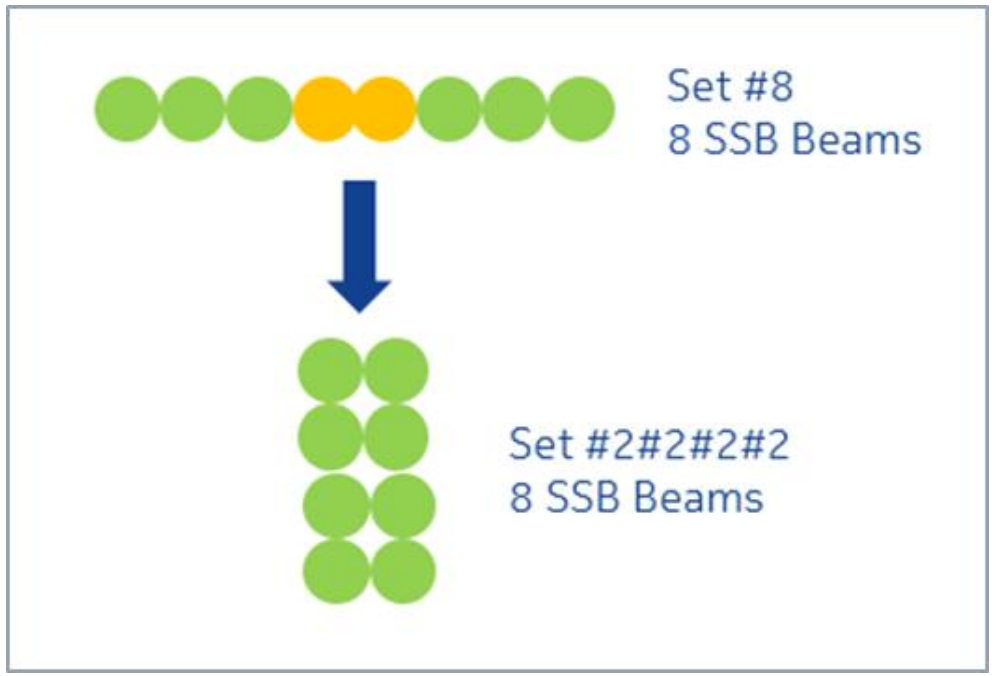
OSS based Antenna Optimization (beamset configuration optimization)

- OSS beamforming counters can be used for analysis. There are 3 categories that can bring useful information and support the direct optimization of the antenna
- Angular user distribution. These counters are related to UE distribution in each beam. They can help to select the most appropriate beamset configurations or to rotate in tilt and azimuth the antenna towards higher density user distributions
- Beam-switching (including the beam before cell HO)
- Time on beams

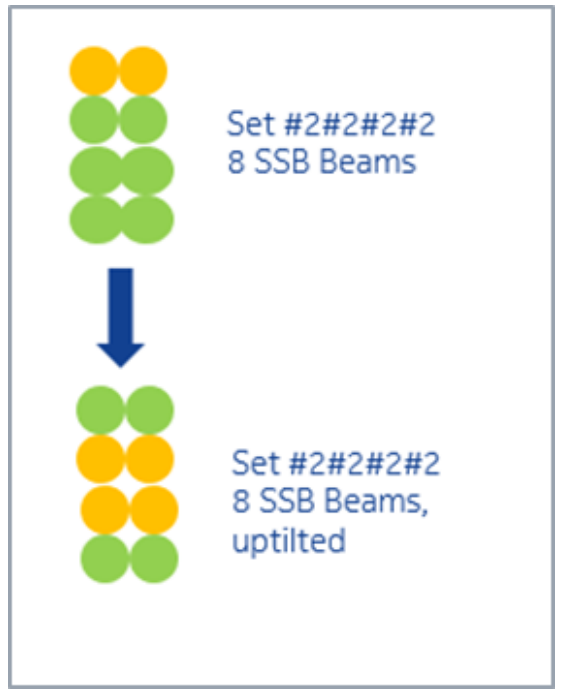
Angular distribution of UEs

This analysis can for example be used to decide if a different beamset should be used or if the antenna should be rotated to give better coverage

Example: If digital beam set #8 is configured, and it is mainly the two middle beams which are used, and manual map inspection shows single high-rise building in the direction of the two middle beams, it should be considered to change to beam set #2#2#2#2



Example: If #2#2#2#2 is already in use to cover a high-rise building, but only the top row of beams is used and visual inspection indicates that the top floors may not have coverage, it can be considered to up-tilt the antenna in order to also provide coverage to the top floors



Angular distribution of UEs

The following counters can be considered in the analysis

UEs_PER_BEAM_ID_xx: These counters seems to have had a bug in 5G19A because the values were much too high. The values in 5G19B onwards look reasonable

RA_ATT_CONT_A_PRMBL_SSB_xx: Typically the ghost preambles far outnumber the real preambles, so these counters cannot be used to understand how the UEs are distributed

RA_ATT_DED_PRMBL_SSB_xx: Dedicated preambles cannot be used for initial access before SRAN20C, so not yet any experience with these counters

DL_SERV_BEAM_ID_BEAM_xx: Good counter to understand how the active UEs are distributed

UL_SERV_BEAM_ID_BEAM_xx: If proactive UL scheduling is enabled (particularly if the scheduling interval is short), the proactive UL schedulings are the most frequent scheduling type in the network. This means these counters are well suited to understand how the connected UEs are distributed

Mobility of UEs in the cell – beam change counters

Counter id	Counter name
M55305C03001..64	BEAM_CHANGE_0..7_0..7
M55305C05001..64	BEAM_CHANGE_TARGET_BIN1..64

If the number of SSB beams does not exceed 8, the BEAM_CHANGE_0..7_0..7 counters will form a matrix, such that the number of beam change attempts between any two SSB beams is counted

If the number of SSB beams exceed 8, the BEAM_CHANGE_0..7_0..7 counters identify the source beam and the BEAM_CHANGE_TARGET_BIN1..64 counters will identify the target beam. It is therefore not possible to see the number of beam changes between specific pair of beams

If refined beams are in use, a beam change from one refined beam to another refined beam within the same SSB beam will be counted as a change with same SSB, e.g. BEAM_CHANGE_3_3

Mobility of UEs beyond the cell

These counters indicate the distribution of the serving Beam ID which UE was using before leaving the cell by HO

If refined beams are not used, the counters indicate the SSB beam. If refined beams are used, the counters indicate the refined beam

Counter id	Counter name
M55305C02001..64	SERV_BEAM_ID_HO_00..63

Mobility of UEs beyond the cell

Counter id	Counter name
M55305C11001..13	TIME_BEAM_BIN_1..13
M55305C03001..64	BEAM_CHANGE_0..7_0..7
M55114C00011	AVE_NSA_USER_HOLDING_TIME

Knowledge of UE mobility is often useful

Such knowledge can be generated e.g. by looking at KPIs such as handovers per minute

Analyzing the time spent on each beam can potentially give better resolution than looking at the whole sector like handover counters do

Time on beam can either be taken from the beam time distribution counters (distribution is directly available, an average can be estimated) or by dividing the number of beam changes with the connection duration (only the average is available

PI WORKS

Features & Use Cases

Features and Parameter Recommendations

Beamforming, su-MIMO & m-MIMO parameter and features



Beamforming
eatures&Paramete

Peak Throughput Impact

- Increasing the number of beams increases the requirement for
  - slots dedicated to the SS/PBCH
  - slots dedicated to the Tracking Reference Signal (TRS)
- The rel. 5G 20 A implementation allows PDSCH data to be scheduled during slots allocated to the SS/PBCH and TRS
- This can mitigate the impact on the peak throughput capability
- In contrast, increasing the number of beams can increase the average cell throughput
- Field Trials required to quantify the balance between peak and average throughputs

Impact before 5GC002534

Number of Beams	Number of SS/PBCH Slots	Peak Throughput	Impact
1	1	551 Mbps	Benchmark
2	1	546 Mbps	-1 %
4	2	525 Mbps	-5 %
6	3	506 Mbps	-8 %
8	4	475 Mbps	-14 %

Improvements with 5GC002534

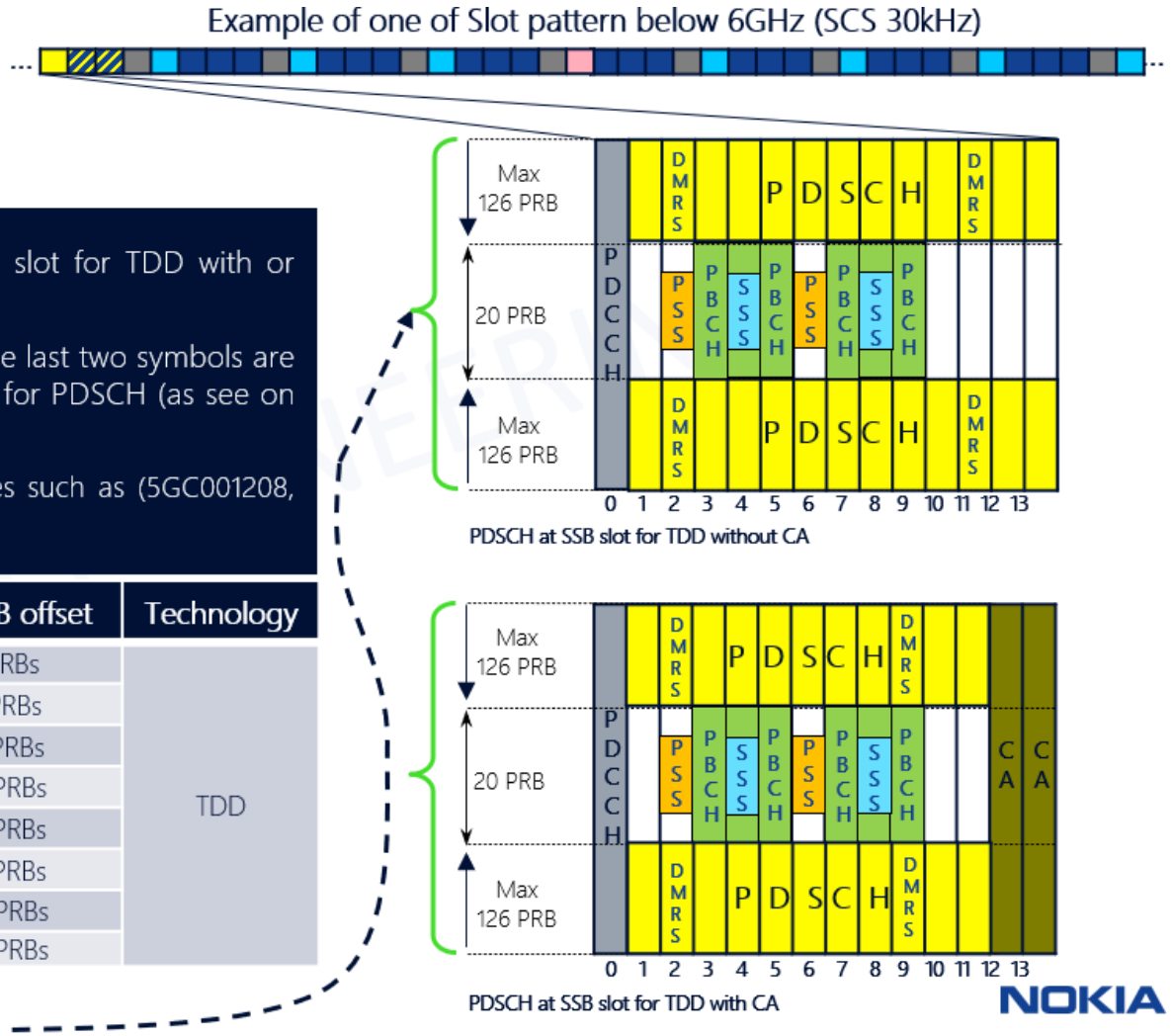
System	Throughput improvement (%)
TDD	2.5%
FDD	3.5%
DSS	5.3%

Feature available for TDD/FDD and DSS

Example PDSCH at SSB slot for TDD

- 5GC002534 also allows to scheduling PDSCH at SSB slot for TDD with or without CSI-RS for Channel Acquisition (CA)
- In case of CA, symbols 1-11 are used for PDSCH and the last two symbols are used for CA. In case of no CA, symbols 1-13 are used for PDSCH (as see on the right side)
- PDCCH is scheduled in symbol 0 in the frame structures such as (5GC001208, 5GC0001070, 5GC001116-0ms or 5GC001116-3ms)

Frequency	SCS	Bandwidth	#PRBs	SSB PRB offset	Technology
Below 6GHz	30 kHz	20 MHz	51 PRBs	31 PRBs	TDD
		40 MHz	106 PRBs	86 PRBs	
		50 MHz	133 PRBs	113 PRBs	
		60 MHz	162 PRBs	142 PRBs	
		70 MHz	189 PRBs	169 PRBs	
		80 MHz	217 PRBs	197 PRBs	
		90 MHz	245 PRBs	225 PRBs	
		100 MHz	273 PRBs	253 PRBs	



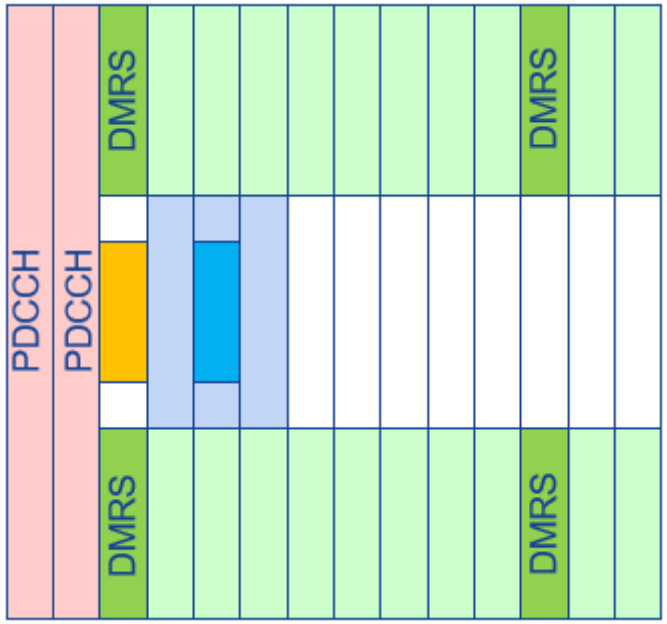
Introduction

- With 5G20B, 5GC002534 'PDSCH at SSB Slots' allows the PDSCH to be scheduled within slots used by the SS/PBCH (not those used by the TRS)
  - restricted to CPRI radio units
  - maximum of rank 2 is allocated to the PDSCH within the SS/PBCH slots
  - not supported in combination with beam refinement, i.e. *nrBtsBeamRefinementP2* must be set to 'False'

- With 5G21A, 5GC002538 'PDSCH Extension in SSB and TRS slots' provides enhanced support:
  - extended to eCPRI radio units
  - maximum of rank 4 is allocated to the PDSCH within the SS/PBCH slots
  - can be supported in combination with beam refinement, i.e. nrBtsBeamRefinementP2 can be set to 'True'

Resource Allocation

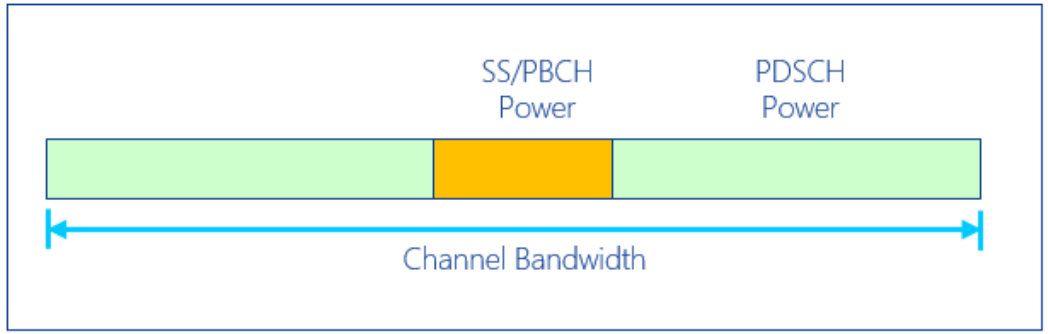
- The Resource Blocks above and below the SS/PBCH can be allocated to a single UE
  - non-contiguous Resource Allocation provided using a bitmap of Resource Block Groups
- Resource allocation is not Rate Matched around the SS/PBCH so Resource Blocks within the bandwidth of the SS/PBCH are not allocated to the PDSCH



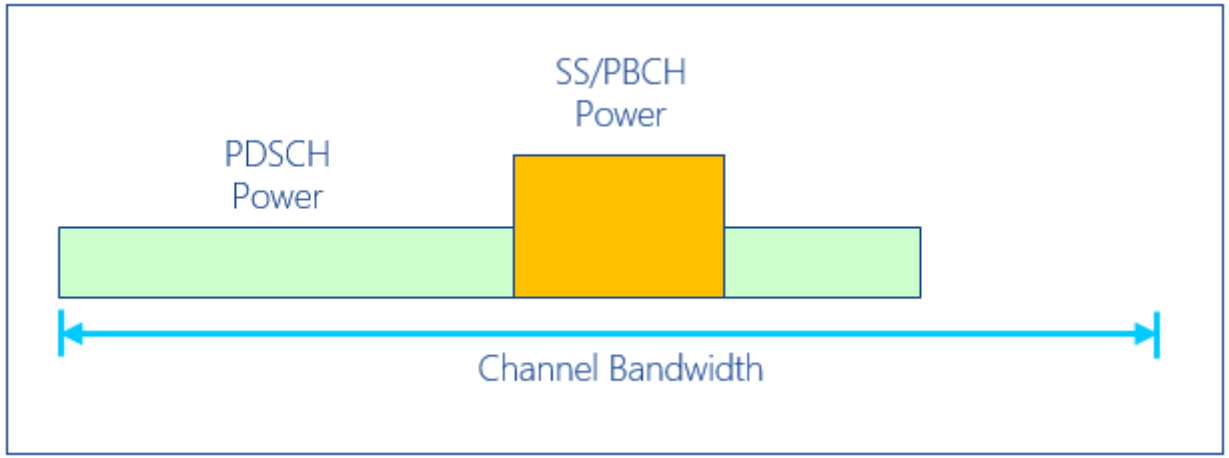
SS/PBCH Transmit Power Impact

- If the SS/PBCH transmit power is boosted then there may be insufficient power to schedule the PDSCH within all remaining PRB
- In this case, SS/PBCH transmit power boosting can have a negative impact upon PDSCH throughput

- SS/PBCH Power is NOT Boosted
- All available PRB scheduled for PDSCH



- SS/PBCH Power is Boosted
- Reduced PRB scheduled for PDSCH



Beamforming Introduction

Nokia Beamforming Evolution

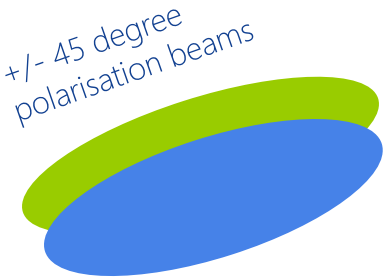


Introduction

- Beamforming is currently applicable to:
  - Active Antenna (mMIMO 32T32R, 64T64R, 2T2R)
  - Passive Antenna with Remote Radio Head (RRH) (8T8R)
  - TDD (provides channel reciprocity)

- Active Antenna are able to generate multiple rows of beams
- Passive Antenna limited to a single row of beams
- Both solutions can theoretically generate an infinite number of beam positions
- Beams become more directional with increased number of antenna elements, i.e. higher antenna gains

- Active and Passive antenna use +/- 45-degree polarisations
- Separate beams are generated for each polarisation



- When talking about beams, we may refer to up to 8 'beams' for example
  - corresponds to up to 16 polarisation beams

- A single MIMO layer uses a single polarisation beam
- Rank 2 transmission requires 2 polarisation beams

} Can be supported using a single 'beam'

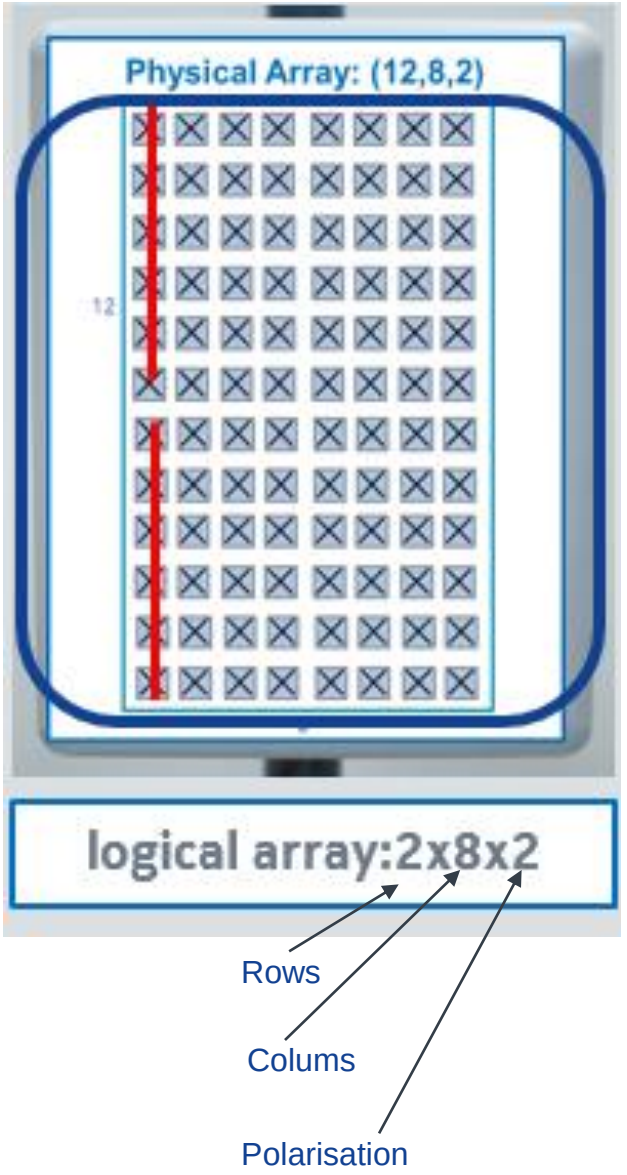
- Rank 3 transmission requires 3 polarisation beams
- Rank 4 transmission requires 4 polarisation beams

} Requires 2 'beams'

Example Active Antenna (FR1)

- Example Active Antenna shown
  - 32T32R
  - 16 TRX for +45˚ arranged as {2 rows × 8 columns}
  - 16 TRX for -45˚ arranged as {2 rows × 8 columns}
  - 192 antenna elements
  - each TRX connected to 6 antenna elements, i.e. 192 /32 = 6

- Increased number of antenna elements increases antenna gain
- Increased TRX improves beamforming control
- Requirement to consider physical size, weight and power consumption



Example Active Antenna (FR2)

- Frequency Range 2 uses Analogue rather than Digital Beamforming
  - single TRX per polarisation beam
  - 2T2R solution supports 2x2 MIMO
- Increased number of antenna elements provide higher gain to help compensate for the higher propagation losses
- Multiple panel solutions available to provide 180˚ beamwidth
- Analogue beamforming allows only 1 beam at a time so Multi-User MIMO requires a modified solution with increased number of TRX (not yet supported)

AirScale Small Cells MAAs

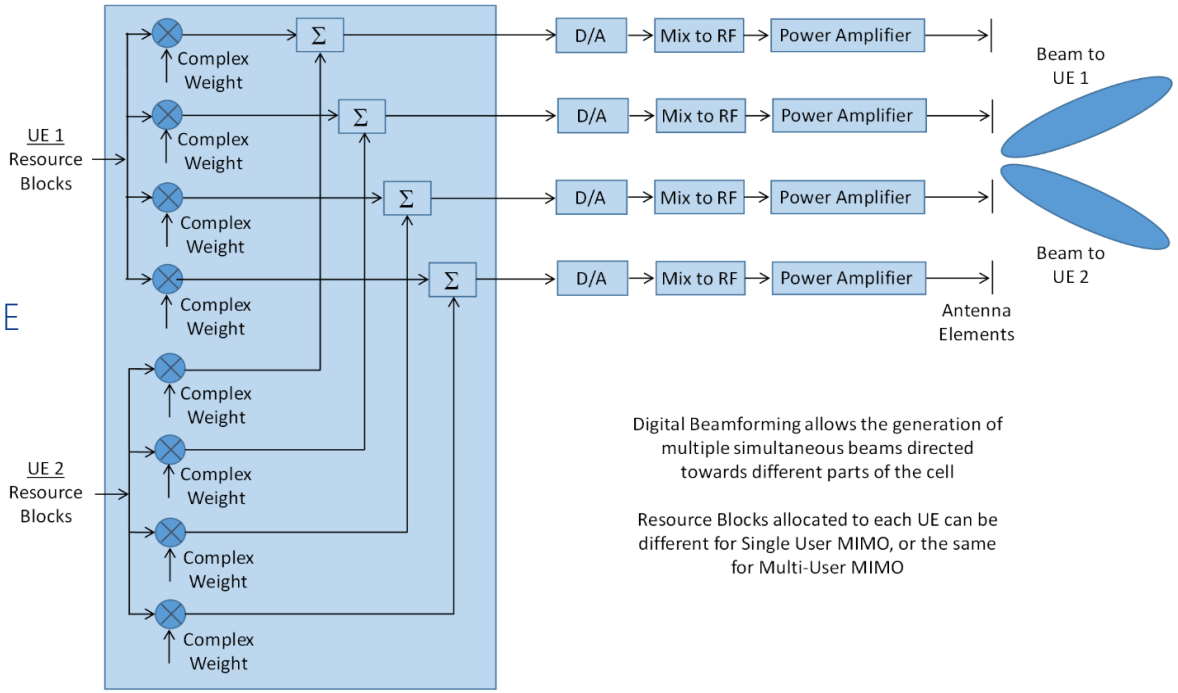
- AirScale MAA 2T2R 512 AE n257 (AEUA)
- AirScale MAA 2T2R 512 AE n257 (AEUB)
- AirScale MAA 2T2R 512 AE n257 (AEUF)
- AirScale MAA 2T2R 512 AE n260 (AEWB)
- AirScale MAA 2T2R 512 AE n260 (AEWF)
- AirScale MAA 2*2T2R 256 AE (AEUD)
- AirScale MAA 2*2T2R 256 AE (AEUE)
- AirScale MAA 2*2T2R 256 AE n260 (AEWD)
- AirScale MAA 2*2T2R 256 AE n260 (AEWE)



Digital Beamforming & Analogue Beamforming

Digital Beamforming

- Applicable to FR1
- Beamforming weights applied prior to D/A
- Flexibility in terms applying different weights to different Resource Blocks
- Allows multiple simultaneous beams to serve different MIMO layers or different UE
- Requires relatively high number of TRX

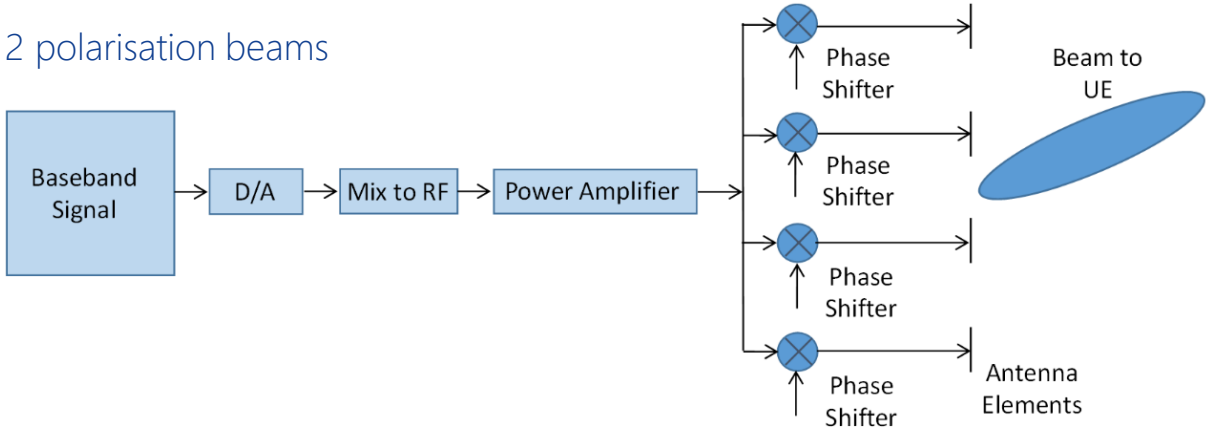


Digital Beamforming allows the generation of multiple simultaneous beams directed towards different parts of the cell

Resource Blocks allocated to each UE can be different for Single User MIMO, or the same for Multi-User MIMO

Analogue Beamforming

- Applicable to FR2
- Beamforming weights applied after D/A
- Beamforming weights applied to the total set of Resource Blocks so one beam at a time
- Limited to 2x2 MIMO served by 2 polarisation beams
- Requires reduced TRX



Digital Beamforming & Analogue Beamforming

Key subfeatures in Digital Beamforming

5GC000533-A: Beamforming on up to 8 SSB beams

5GC000533-B: Beam tracking on single SSB beam

5GC000533-C: Beam refinement and reporting of second-best beam

(descoped from original feature and introduced in 5G19A with 5GC001942 feature)

5GC000533-E: Beam recovery and PM Counters

Key subfeatures in Analogue Beamforming

5GC000535-A: SS block burst set beamforming control

5GC000535-B: Analog BF with basic configuration: initial access and data transfer

5GC000535-C: Analog BF with multiple carriers

5GC000535-D: Beam tracking

5GC000535-E: Beam recovery through PRACH

5GC000535-F: Analog BF with multiple GoB patterns

5GC000535
Analog Beamforming
5GC001943 Spillover analog Beamforming (5G19A)

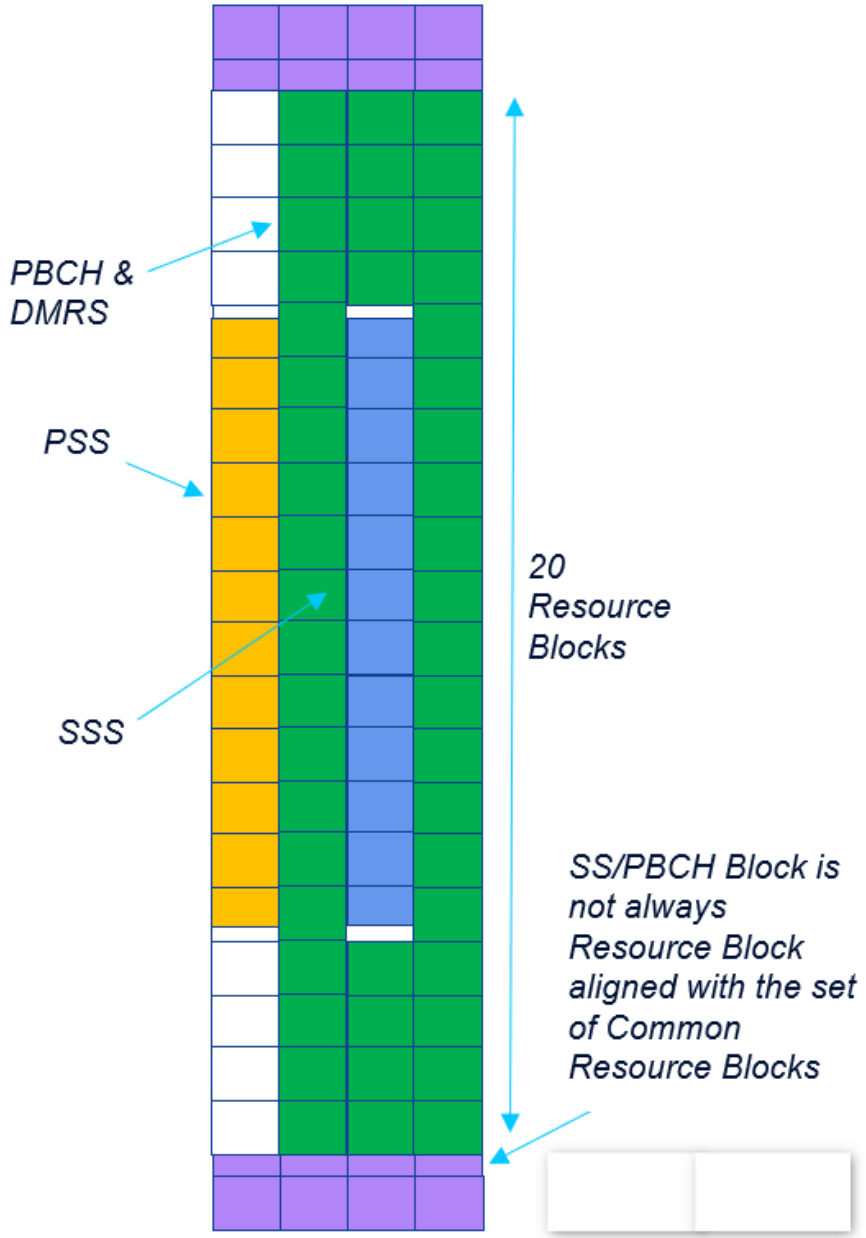
SS/PBCH based Beamforming

SS/PBCH Block

- An SS/PBCH Block allows;
  - synchronisation in time and frequency
  - acquisition of Master Information Block (MIB)
  - measurement of RSRP/RSRQ
- SS/PBCH beamforming creates a beam for each SS/PBCH and its associated Tracking Reference Signal (TRS)
- A UE selects the best SS/PBCH, and then the corresponding beam is used for uplink reception and downlink transmission
- SS/PBCH beamforming improves both uplink and downlink coverage (except Long PUCCH in some cases)

Parameter	Object	Range	Default
ssPbchBlockPower	NRCELL	-60 to 50, step 1 dBm	-

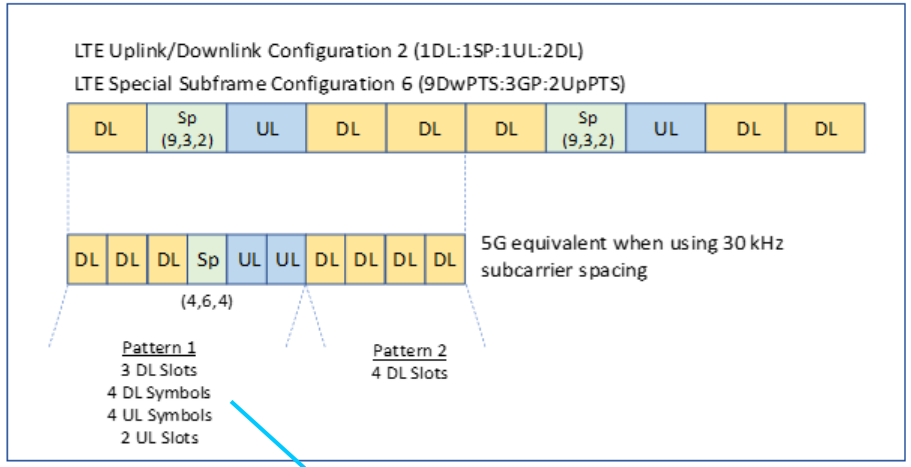
Power per Resource Element 'per Polarisation Beam' (when beamforming is enabled)



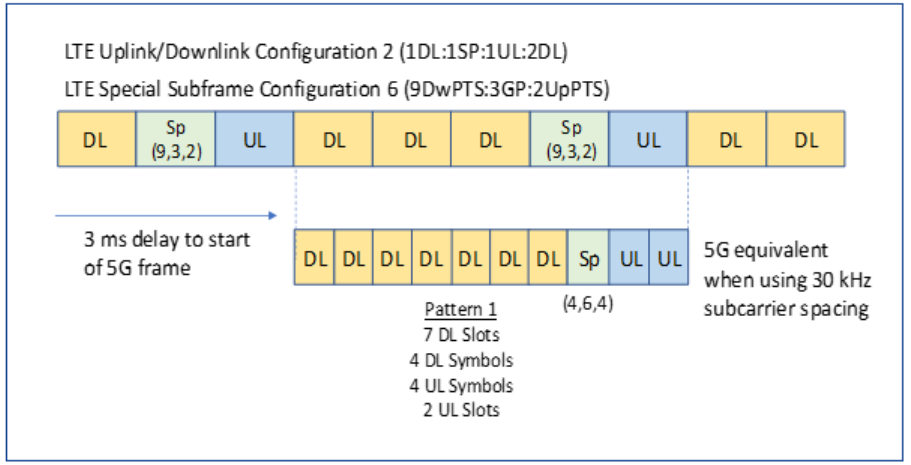
SS/PBCH Transmissions

- Each beam requires a time multiplexed SS/PBCH (SSB)
- There can be up to 2 SSB per slot so 4 consecutive DL slots are required to support 8 beams
  - requires the 3 ms time shift when using the tdLTE uplink/downlink slot pattern

***lteToNrFrameShift** = 0 ms*



***lteToNrFrameShift** = 3 ms*



	Symbol														
	Slot	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Frame 0	0			SSB	SSB	SSB	SSB			SSB	SSB	SSB	SSB		
	1			SSB	SSB	SSB	SSB			SSB	SSB	SSB	SSB		
	2			SSB	SSB	SSB	SSB			SSB	SSB	SSB	SSB		
	3						Guard	Guard	Guard	Guard	Guard	Uplink	Uplink	Uplink	Uplink
	4	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink
	5	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink	Uplink
	6														

Best Beam Reporting

- The SS/PBCH are beamformed according to the parameter design (NRCELL – beamSet – basicBeamSet)
- The UE measures the RSRP of each beam and reports the 'top 2'
- The BTS subsequently applies beamforming to the transfer of uplink and downlink data on the PUSCH/PDSCH using the selected beam(s)

- Up to 6 beams for Passive Antenna and 8 beams for Active Antenna (Frequency Range 1)

```
reportConfigId 2,
resourcesForChannelMeasurement 10,
reportConfigType periodic :
    reportSlotConfig slots320 : 235,
    pucch-CSI-ResourceList
        uplinkBandwidthPartId 1,
        pucch-Resource 12
   ,
reportQuantity ssb-Index-RSRP : NULL,
timeRestrictionForChannelMeasurements notConfigured,
timeRestrictionForInterferenceMeasurements notConfigured,
groupBasedBeamReporting disabled :
    nrofReportedRS n2
```

UE report sent on the PUCCH every 320 slots

Reporting of best SS/PBCH beam and its RSRP

Top 2 SS/PBCH beams can be reported

Grid of Beams
Example beam Patterns



Set #6
6 SSB Beams



Set #4#4
8 SSB Beams



Set #5#3
8 SSB Beams

Beamforming helps to improve RSRP (cell coverage)

Beam Set Selection

- Requirement to select beamforming configuration
  - number and arrangement of beams
- Digital Beamforming is applicable to Frequency Range 1
- Analogue Beamforming is applicable to Frequency Range 2

It is necessary to verify that the Slot Format has sufficient downlink slots to accommodate the target number of SS/PBCH, e.g. beamSet_4_4 requires 8 SS/PBCH so it requires 4 downlink slots at the start of the radio frame

Parameter	Object	Range		Default
basicBeamSet	NRCELL	1	beamSet_1	beamSet_1 (1)
		2	beamSet_2	
		3	beamSet_4	
		4	beamSet_8	
		5	beamSet_4_4	
		6	beamSet_5_3	
		7	beamSet_6_2	
		9	beamSet_2_2_2_2	
		10	beamSet_3_3_2_square	
		11	beamSet_3_3_2_azimuth	
		16	beamSet_6	
		22	beamSet_2_2	
		24	beamSet_2SS	
		25	beamSet_4SS	
		31	beamSet_3_1	
		33	beamSet_3_3	
		42	beamSet_4_2	
		51	beamSet_5_1	
		100	beamSetAbf_1A	
		101	beamSetAbf_32A	
		102	beamSetAbf_32B	
		103	beamSetAbf_32C	
		104	beamSetFR2_16_8_8_1P	
		105	beamSetFR2_8_8_8_8_1P	
		106	beamSetFR2_6_6_6_6_6_2_1P	
		201	beamSetFR2_6_6_4_2P	
		301	beamSetFR2_6_2_4P	
		401	beamSetFR2_4_4_2_3P	

Digital Beamforming

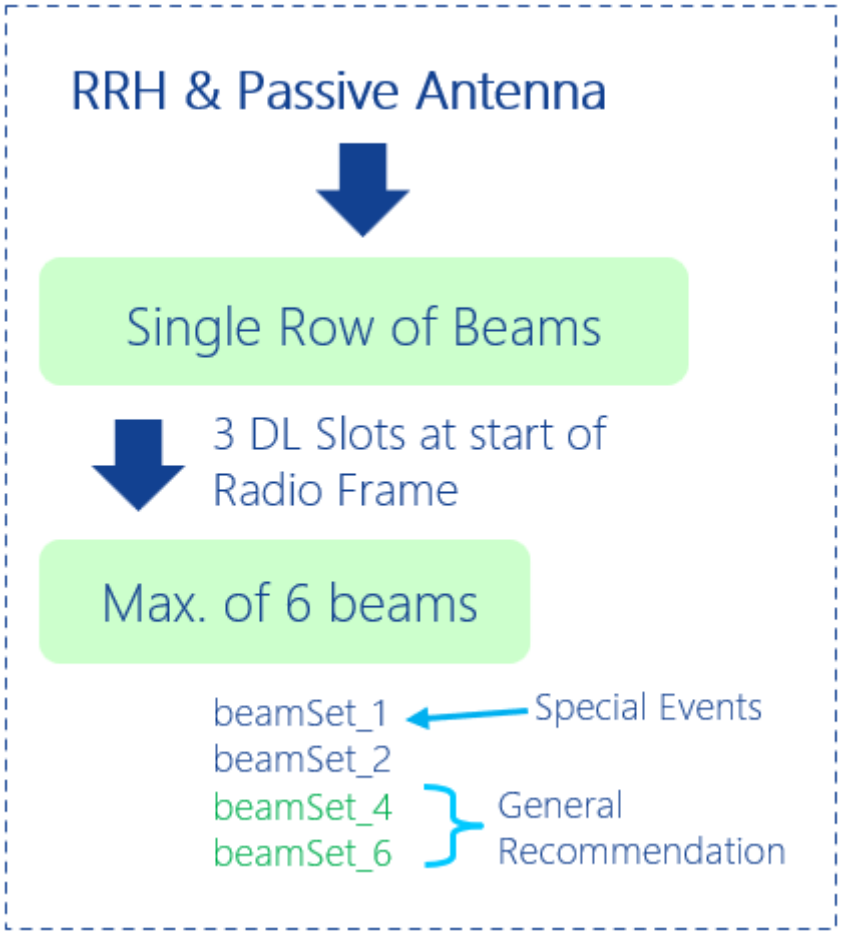
Choice of refined beam patterns

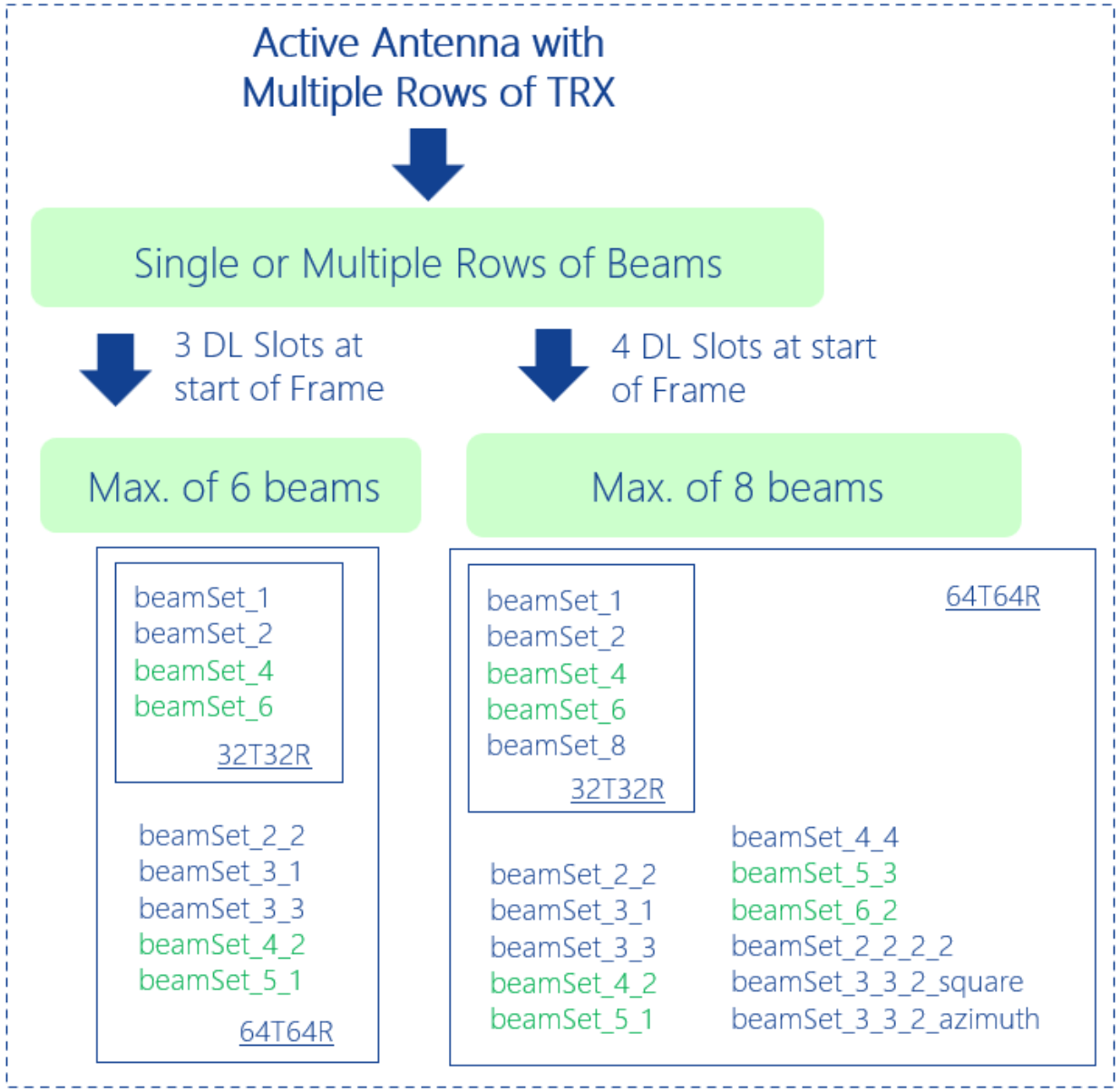
Sub-sector configurations for 8T8R

Analogue Beamforming

Multiple antenna panels

Beam Selection - FR1





32T32R can theoretically support multiple rows of beams but it is not recommended due to having only 2 rows of TRX

Appropriate number of beams depends upon geographic size of cell, i.e. small cells require fewer beams

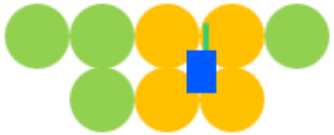
Urban areas may have greater requirement for multiple rows of beams

Beam Set Selection vs 4x4 MIMO

- 4x4 MIMO uses a total of 4 polarisation beams
  - 2 polarisation beams from SS/PBCH beam 'A'
  - 2 polarisation beams from SS/PBCH beam 'B'
- There can be an increased probability of achieving higher rank transmission when using multiple rows of beams
  - can provide the UE with an increased choice of beams



Set #6
6 SSB Beams

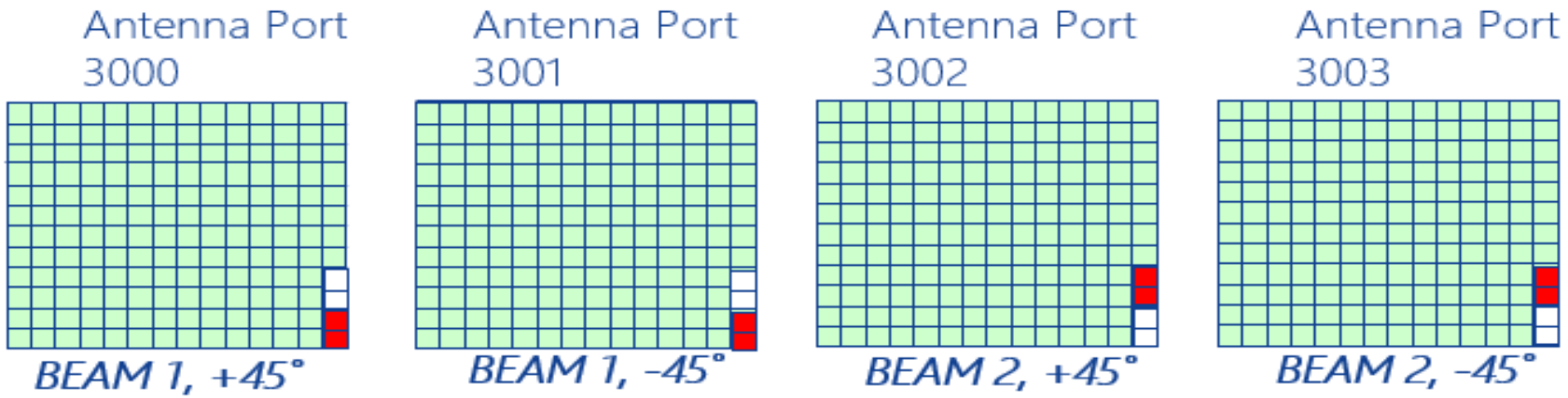


Set #5#3
8 SSB Beams

- This effect has been observed in the field with a project using AEQD (64 TRX) Active Antenna
- In this case, Rank 4 was obtained more readily when using Set #5#3 rather than Set #6

MIMO

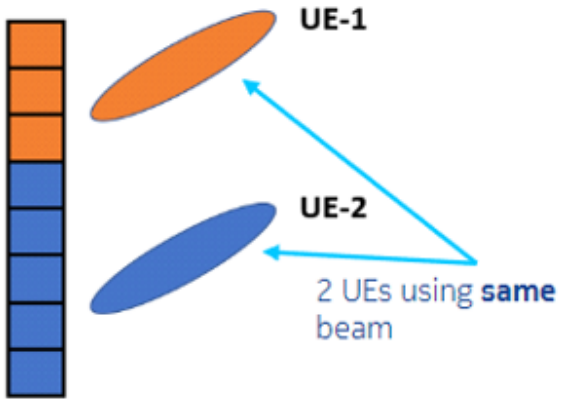
- SS/PBCH based Beamforming relies upon a 'UE specific' CSI Reference Signal for MIMO measurements and reporting
- 'UE specific' CSI RS is beamformed according to the 'top 2' beams reported by the UE
- UE reports CQI, PMI and Rank for MIMO based upon its measurements
- The 'UE specific' CSI RS can create a capacity bottleneck and cause blocking
- Blocking pegs the 'No PUCCH' counter:
  - M55115C00005 SGNB_DU_NSA_ADMISSION_NO_PUCCH



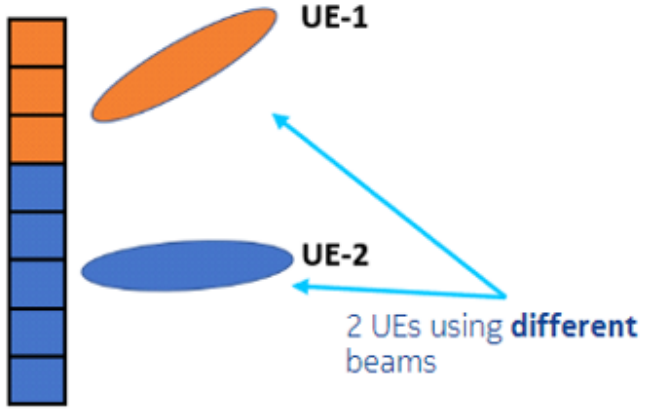

```
csi-MeasConfig setup :
    nzp-CSI-RS-ResourceToAddModList
        nzp-CSI-RS-ResourceId 0,
        resourceMapping
          frequencyDomainAllocation row4 : '001'B,
          nrofPorts p4,
          firstOFDMSymbolInTimeDomain 13,
          cdm-Type fd-CDM2,
          density one : NULL,
          freqBand
            startingRB 0,
            nrofRBs 108
         ,
        powerControlOffset 3,
        powerControlOffsetSS db-3,
        scramblingID 44,
        periodicityAndOffset slots320 : 88
       ,
```

Frequency Domain Beamforming

- Supported by '5GC002537 FDM Allocation Enhancements'
  - requires eCPRI
- Prior to this feature, all UE allocated PRB within the same slot had to use the same beam
- Frequency Domain Beamforming allows frequency multiplexed UE to be allocated different beams



CPRI without Frequency Domain Beamforming



eCPRI with Frequency Domain Beamforming

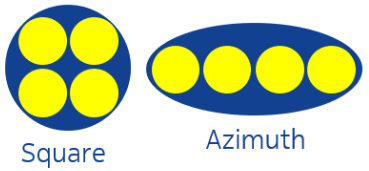
SS/PBCH Beamforming Comments

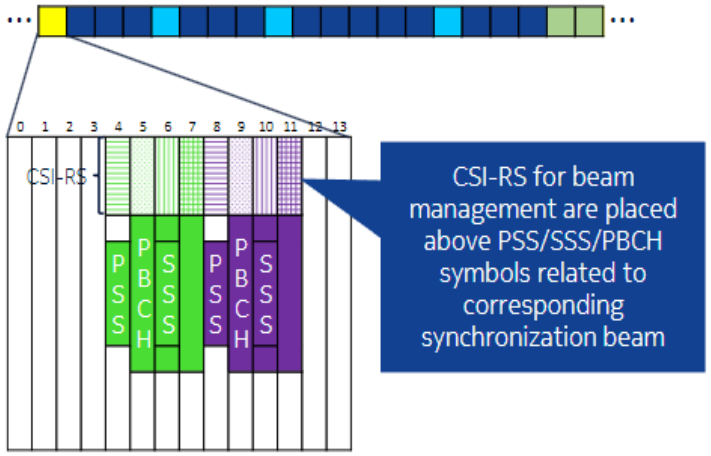
- SS/PBCH beamforming improves both uplink and downlink coverage
- SS/PBCH has not been implemented by all network vendors
  - other vendors may use SS/PBCH boosting to compensate (only increases downlink RSRP)
  - or other vendors may offer larger antenna arrays with higher gain

- Configuring a single SS/PBCH beam at Special Events helps to increase capacity at locations where increased coverage is not required
  - avoids the requirement for 'Best SSB Index' reporting on the PUCCH and reduces SS/PBCH overhead

Beam Refinement

- ‘Beam Refinement’ can be used in addition to SS/PBCH Beamforming for Active Antenna in FR1
- The Channel State Information Reference Signal (CSI RS) is broadcast using more directional beams
  - 4 refined beams per SS/PBCH beam
  - up to 32 beams available for selection
- CSI RS are frequency multiplexed with the SS/PBCH
- UE reports the CSI RS identity of the best 1 or 2 beams





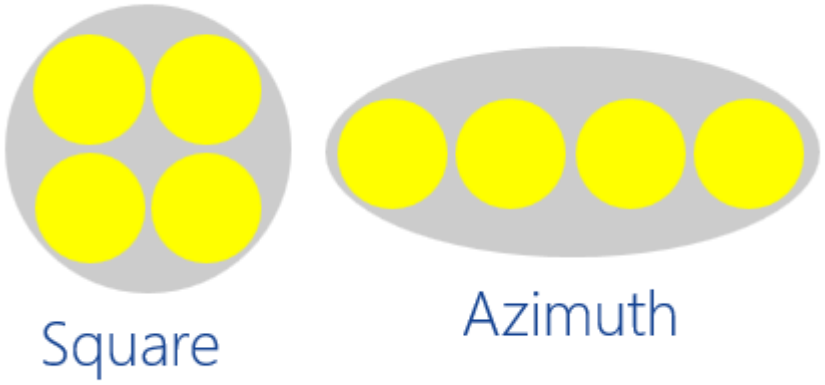

```
reportConfigId 2,
resourcesForChannelMeasurement 10,
reportConfigType periodic :
    reportSlotConfig slots320 : 84,
    pucch-CSI-ResourceList
        uplinkBandwidthPartId 1,
        pucch-Resource 12
   ,
reportQuantity cri-RSRP : NULL,
timeRestrictionForChannelMeasurements notConfigured,
timeRestrictionForInterferenceMeasurements notConfigured,
groupBasedBeamReporting disabled :
    nrofReportedRS n2
```

Reporting of best CSI RS beam and its RSRP

```
nzp-CSI-ResourceSetId 2,
nzp-CSI-RS-Resources
  34,
  35,
  36,
  37,
  38,
  39,
  40,
  41,
  42,
  43,
  44,
  45,
  46,
  47,
  48,
  49,
  50,
  51,
  52,
  53,
  54,
  55,
  56,
  57
,
```

Beam Refinement Patterns

- There are always 4 refined beams per SS/PBCH Beam (assuming Beam Refinement is enabled)
- The pattern of refined beams has been implemented as either ‘square’ or ‘azimuth’



		Beam Set	beam refinement	# SSB beams	# refined beams per SSB beam	azimuth opening angle
1 row	o	#1	square	1	4	90° 120°
	o-o	#2	square	2	4	90° 120°
	o-o-o-o	#4	square	4	4	90° 120°
	o-o-o-o-o-o	#6	square	6	4	90° 120°
	o-o-o-o-o-o-o-o	#8	square	8	4	90° 120°
2 rows	o-o-o-o o-o-o-o	#4#4	square / square	8	4	90° 120°
	o-o-o-o-o o-o-o	#5#3	square / azimuth	8	4	90° 120°
	o-o-o-o-o-o o-o	#6#2	square / azimuth	8	4	90° 120°
3 rows	o-o-o o-o-o o-o	#3#3#2	square / square / square azimuth / azimuth / azimuth	8	4	90° 120° 90° 120°
4 rows	o-o o-o o-o o-o	#2#2#2#2	square / square / square / square	8	4	40° 90°

Refined beam pattern per row of beams

Tilting of Beam Pattern (I)

- The *tiltOffset* parameter (5GC001991 'Antenna tilt for 2D GoB') allows a tilt to be applied to the beam pattern
- Applicable to active antenna with 32 or 64 transceivers
- The range of the tilt is hardware specific and also depends upon the beam pattern (see next slides)

Parameter	Object	Range	Default	Description
tiltOffset	NRCELL	-13 to 13 degrees	0 degrees	This parameter indicates the tilt offset that can be applied to 32/64TRX cmWave (FR1) Radios with beamforming capability connected to the system.

- An alarm is triggered if the tilt is set to a value outside the allowed range

Fault ID	Fault name	Reported alarms	
		Alarm ID	Alarm name
1900	RF Module configuring failed	7101	BASE STATION/UNIT AUTONOMOUS RESET NOTIFICATION
4598	Tilt Offset is out of range	7115	BASE STATION INFORMATION

Tilting of Beam Pattern – 32T32R & 64T64R

- The range of permitted *tiltOffset* values for each hardware unit is shown within this table

Table: *Minimum and maximum tilt value of GoB - 32TXRX mMIMO*

Radio name	Antenna panel	Vertical electrical tilt range around pre-tilt and for SLS > 6dB (Tilt_Offset)			
		GoB with one row	GoB with two rows	GoB with three rows	GoB with four rows
AEQN	6x8x2	-2° up to +4°	-3° up to +3°	Not supported	Not supported
AENB	8x8x2	-3° up to +4°	-3° up to 0°	Not supported	Not supported
AEQQ/C	12x8x2	-5° up to +5°	-5° up to +5°	Not supported	Not supported
AEQQ_I	8x8x2	-3° up to +3°	-3° up to 0°	Not supported	Not supported
AEQZ	12x8x2	-1° up to +1°	0°	Not supported	Not supported
AEQV/Y	8x8x2	-3° up to +3°	-3° up to 0°	Not supported	Not supported
AEQQ/C (split mode)	6x8x2	-5° up to +5°	Not supported	Not supported	Not supported
AQQC	8x8x2	-3° up to +3°	-3° up to 0°	Not supported	Not supported
AQQY	8x8x2	-3° up to +3°	-3° up to 0°	Not supported	Not supported
AQQZ	8x8x2	-3° up to +3°	-3° up to 0°	Not supported	Not supported
AQQA/F	12x8x2	-1° up to +1°	0°	Not supported	Not supported
AQHA	12x8x2	-1° up to +1°	0°	Not supported	Not supported
AQQJ	12x8x2	-1° up to +1°	0°	Not supported	Not supported
AEQW	12x8x2	-1° up to +1°	0°	Not supported	Not supported
AQHB	8x8x2	-1° up to +1°	0°	Not supported	Not supported

Table: *Minimum and maximum tilt value of GoB - 64TXRX mMIMO*

Radio name	Antenna panel	Vertical electrical tilt range around pre-tilt and for SLS[1)] > 6dB (Tilt_Offset)			
		GoB with one row	GoB with two rows	GoB with three rows	GoB with four rows
AEQD	8x8x2	-13° up to +13°	-13° up to +12°	-13° up to 0°	0°
AAHF/J[2)]	8x8x2 or 4x8x2	-11° up to +10°	-11° up to +8°	-11° up to 0°	0°
AEQA/B/E/E_I/K/K_I/P/P_I/I	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AEHC	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AQQL	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AEQB/E/E_I/I/J/P/P_I/K/K_I (split mode)[3)]	6x8x2	-3° up to +5°	-3° up to +3°	Not supported	Not supported
AEHB	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AQQE/K	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AEQU	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AQQN/I	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AEQM	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°
AQQS	12x8x2	-6° up to +6°	-6° up to +4°	-6 up to 0°	0°
AQNA	12x8x2	-7° up to +7°	-7° up to +5°	-7 up to 0°	0°

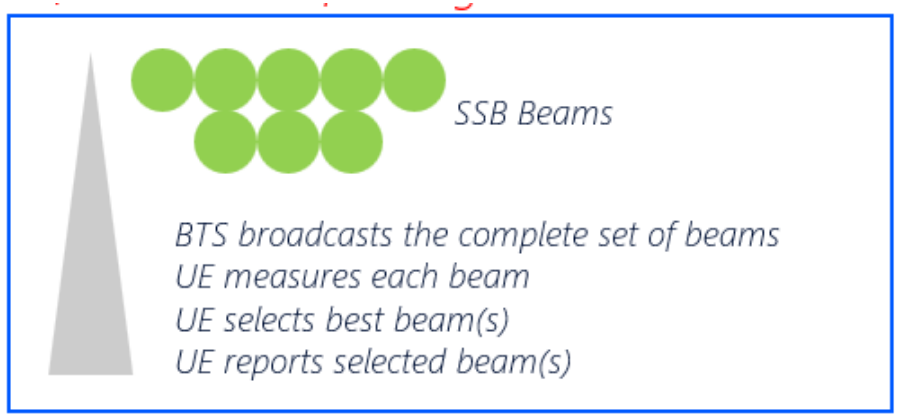
Codebook based Beamforming

5GC002200 4 x ‘8 Port CSI-RS’ Codebook for SU-MIMO

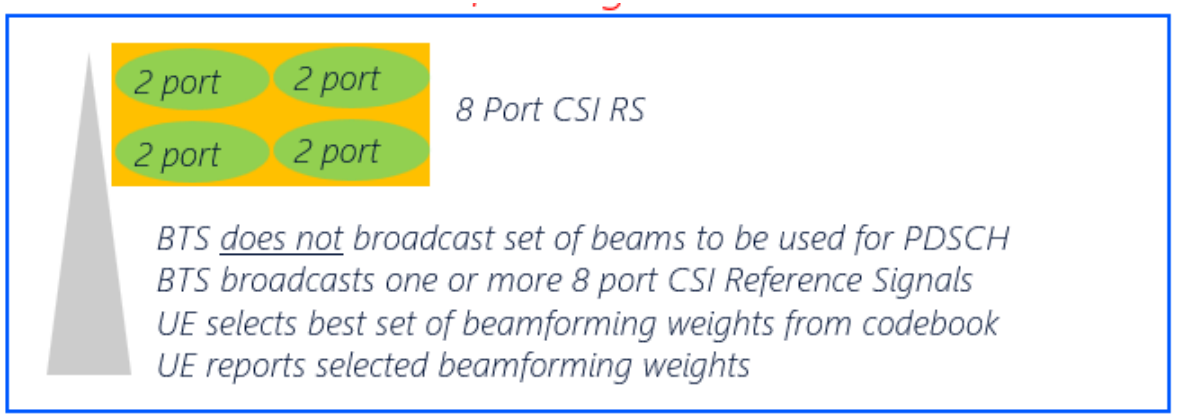
Feature is limited to the PDSCH. Other channels continue to use SS/PBCH based Beamforming

- Supported for both Passive and Active Antenna solutions
- Fundamental change to beamforming solution
- Codebook based Beamforming is more complicated for the UE
  - BTS broadcasts one or more ‘8 Port CSI RS’
  - UE evaluates multiple sets of beamforming weights to determine which would be best
  - UE signals the preferred beamforming weights using the MIMO Precoding Matrix Indicator (PMI)

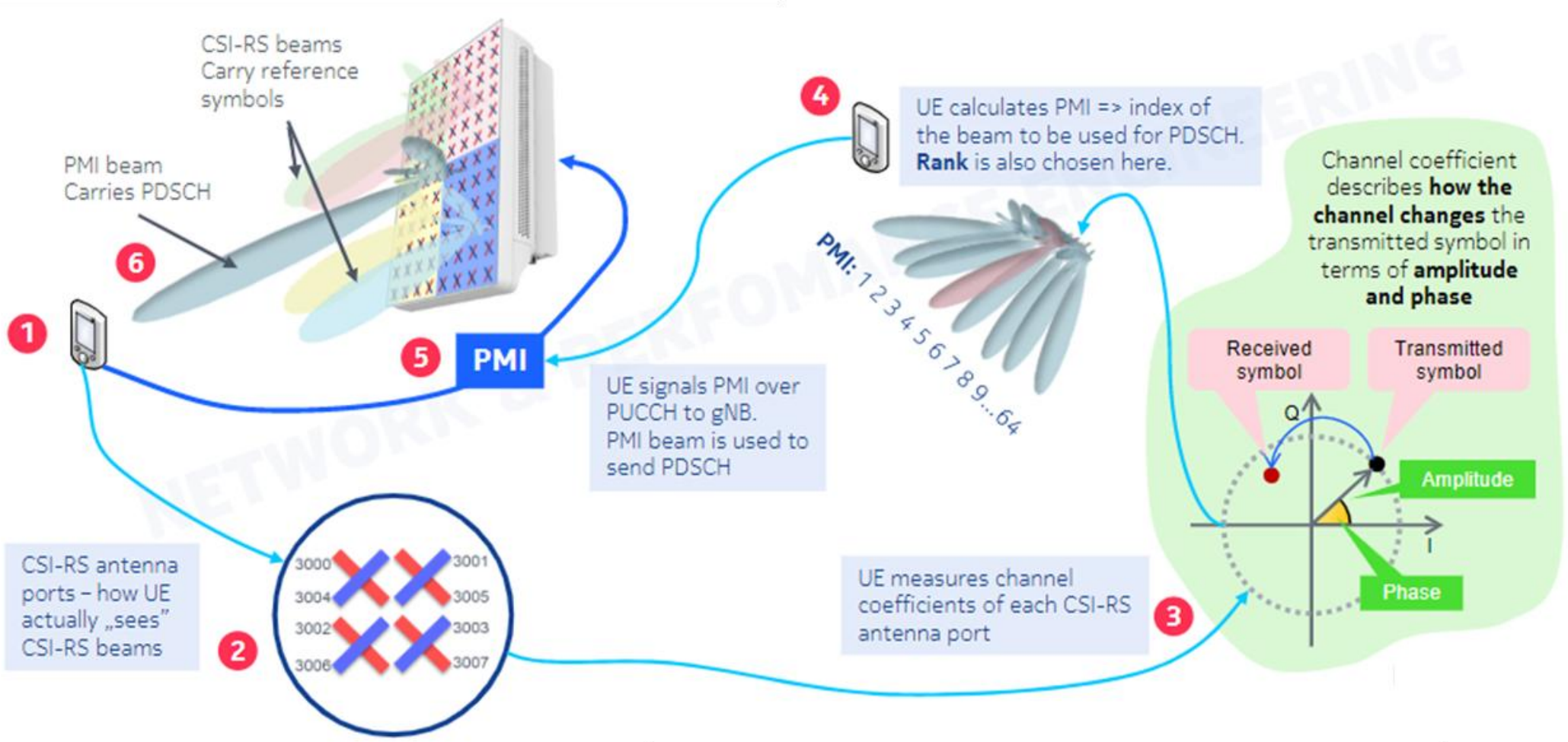
SS/PBCH Beamforming



SS/PBCH Beamforming



Codebook based Beamforming Process



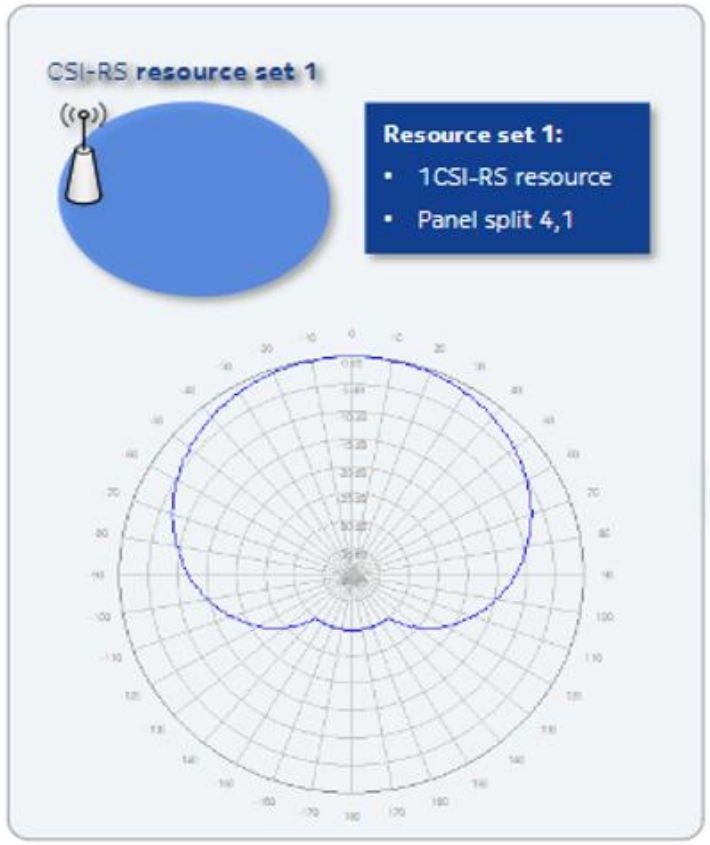
Feature Activation

- Enabled using the NRCELL - *dlMimoMode* parameter

Parameter	Object	Range	Default
dlMimoMode	NRCELL	30 2x2 Closed Loop Spatial Mux 40 4x4 or 4x2 Closed Loop Spatial Mux 50 2x2 Open Loop Spatial Mux 60 4x4 or 4x2 Open Loop Spatial Mux 70 8x4 Closed Loop Spatial Mux	2x2 Closed Loop Spatial Mux (30)

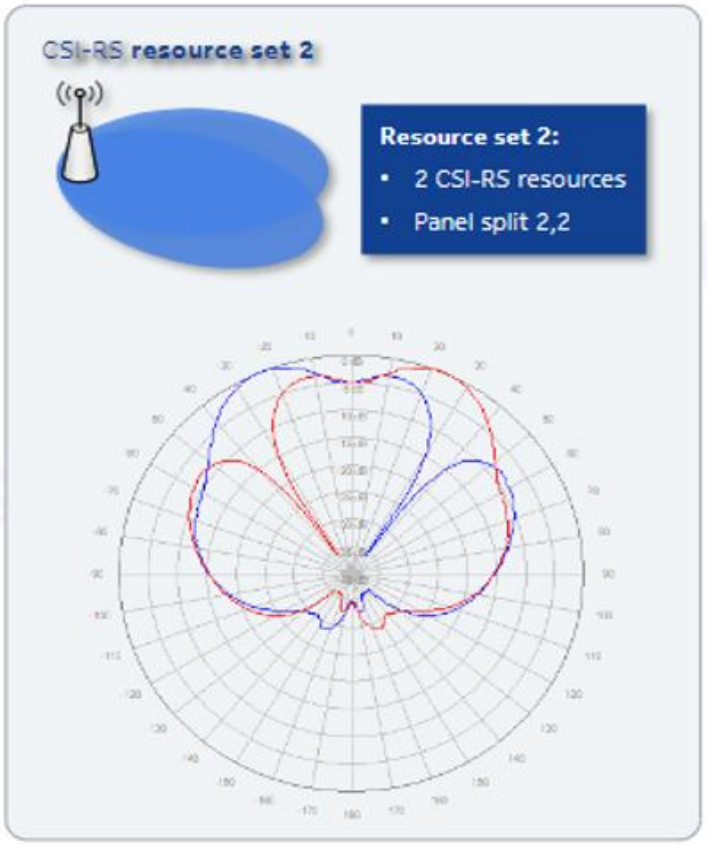
1, 2 or 4 x ‘8 Port CSI Reference Signal’

RRH + Passive Antenna
Active Antenna



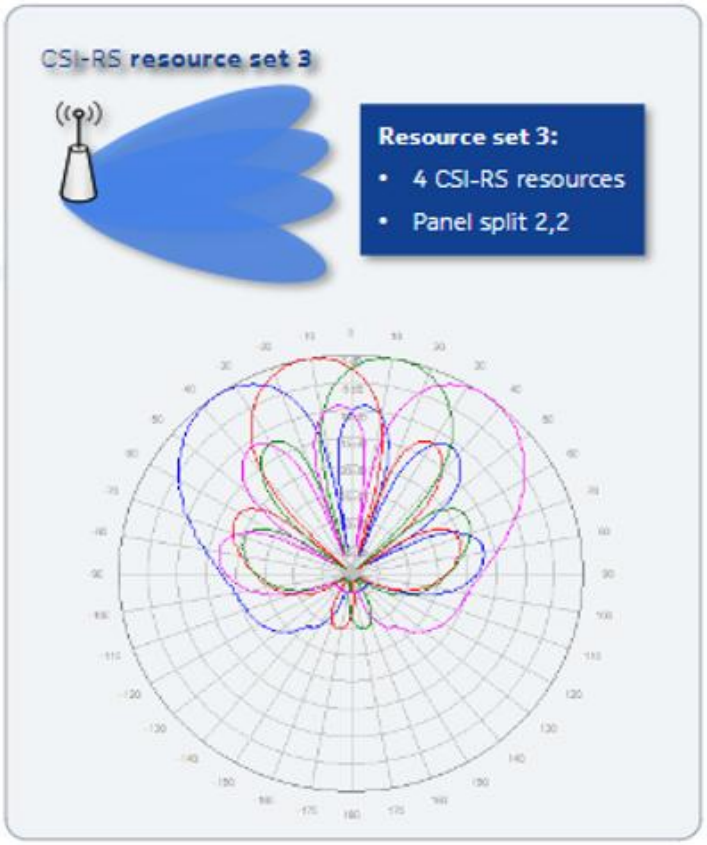
16 beams available

Active Antenna



128 beams available

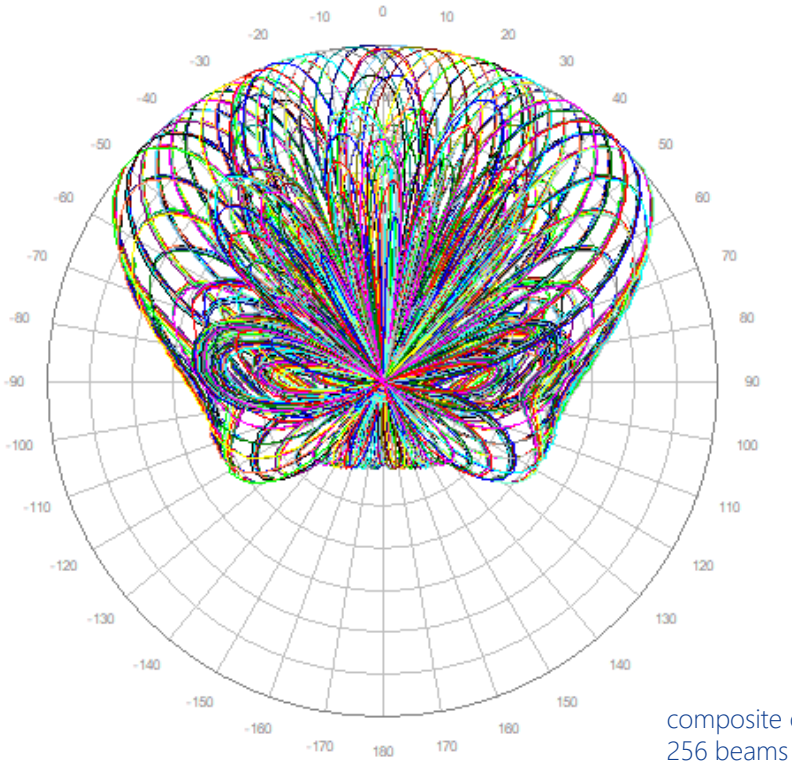
Active Antenna



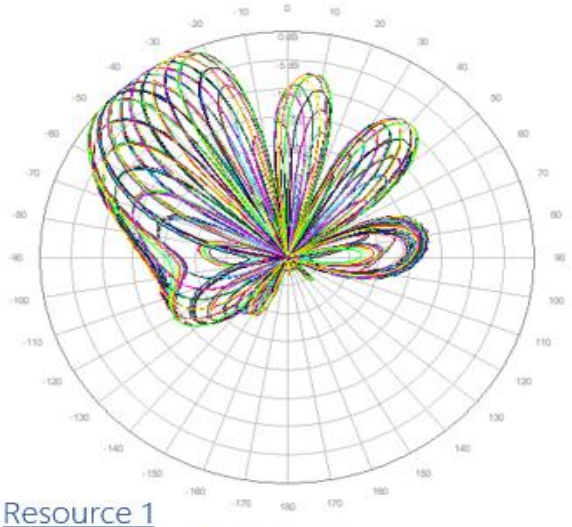
256 beams available

4 CSI RS Resource – Antenna Gain Patterns

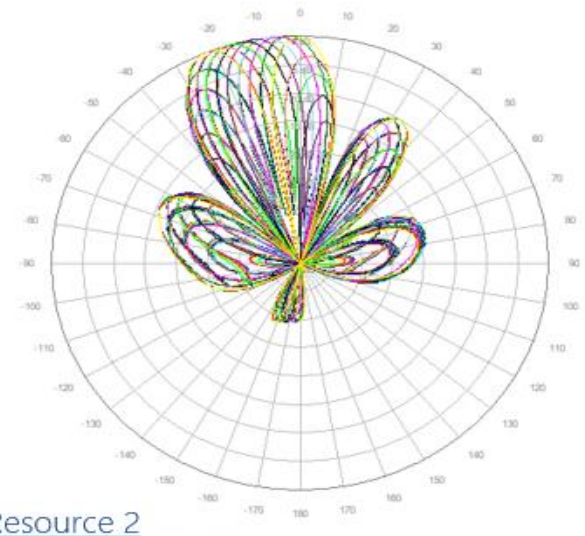
- The 4 CSI RS Resources use the (2,2) configuration which leads to 64 beams per Resources (8x8 grid of beams per Resource)
- The four Resources are arranged from left to right



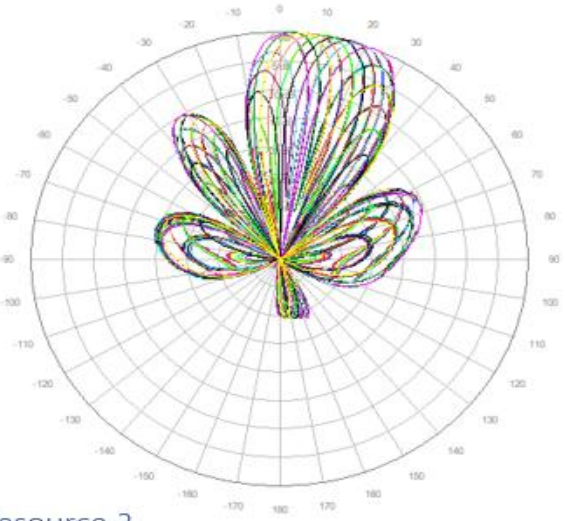
composite of all 256 beams



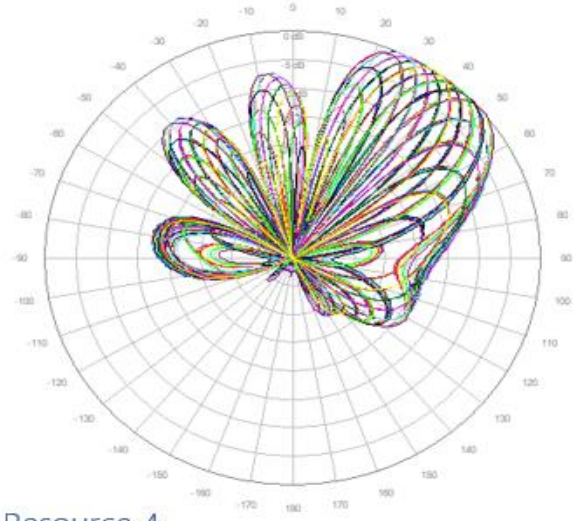
Resource 1
Composite of all 64 beams



Resource 2
Composite of all 64 beams



Resource 3
Composite of all 64 beams

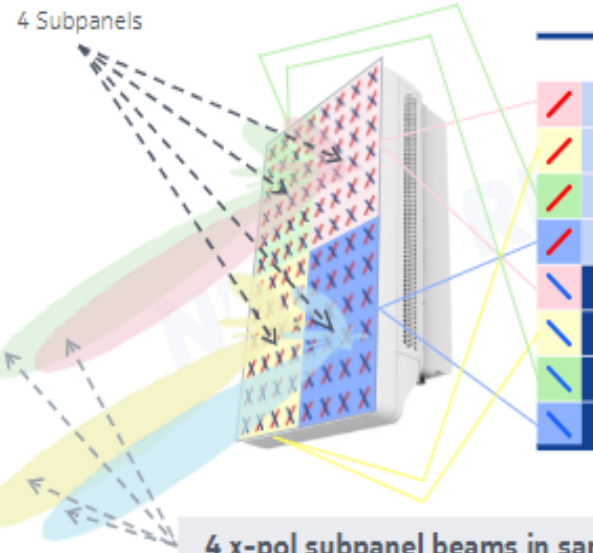


Resource 4
Composite of all 64 beams

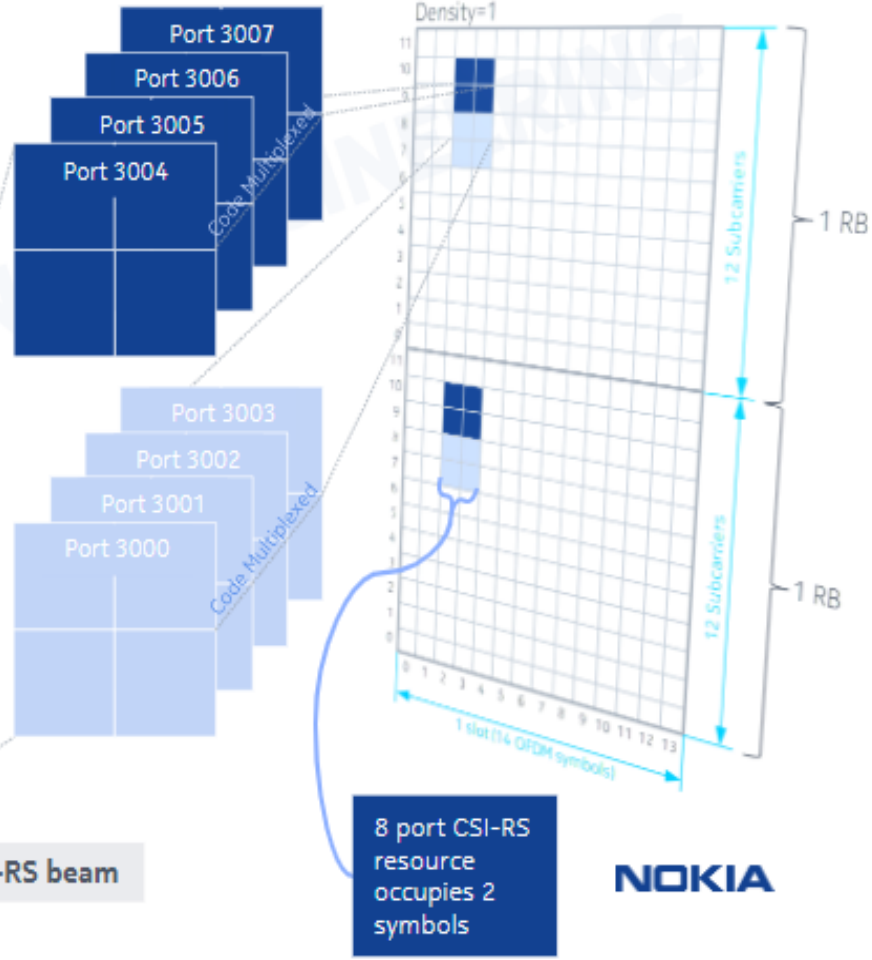
CSI RS Transmission – Active Antenna

Example of one CSI-RS resource mapping to antenna port

- If **dlMimoMode is set to 8x4**, periodic CSI-RS resource for channel state feedback is sent on **dedicated subpanel beams**.
- To distinguish the totally **8 antenna ports**:
  - 4 X-polarization subpanel beams are needed.
  - **Each X-polarization beam serves 2 antenna ports** - one antenna port per polarization.



antenna port #	beam #	polarization
3000	1	+45°
3001	2	+45°
3002	3	+45°
3003	4	+45°
3004	1	-45°
3005	2	-45°
3006	3	-45°
3007	4	-45°



4 x-pol subpanel beams in same direction = 1 CSI-RS resource = 1 CSI-RS beam

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Nokia internal use

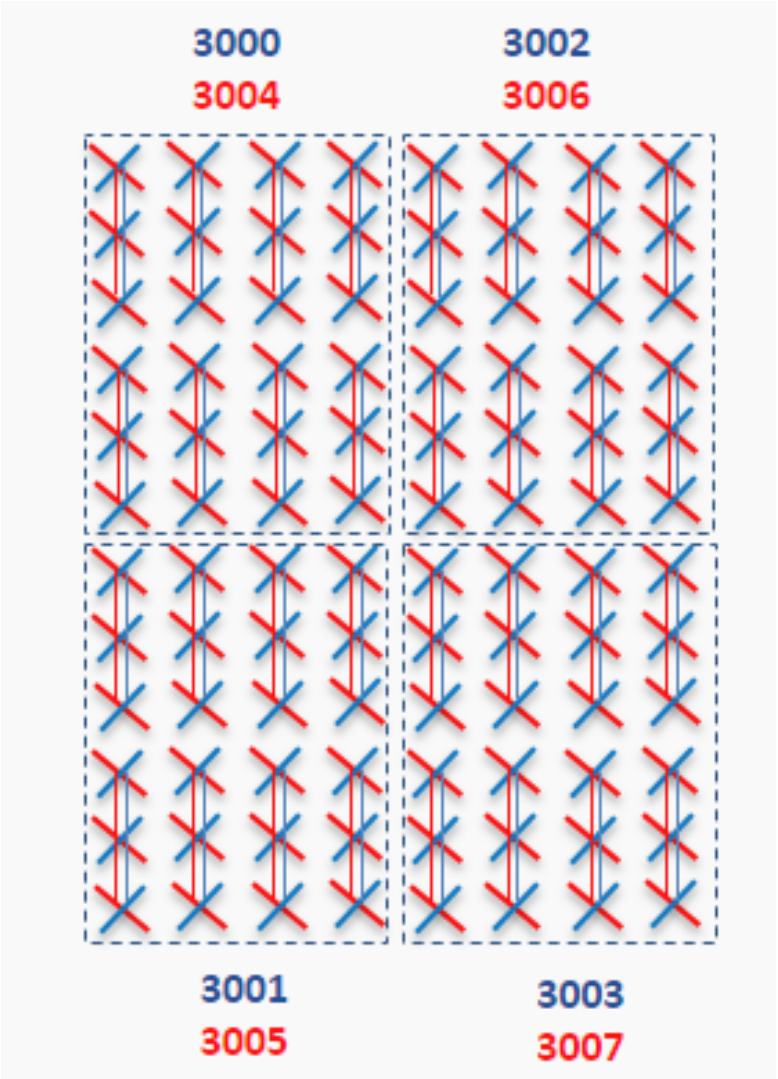
Number of Beams

- 3GPP specifies the following table which provides 2 options for the '8 port CSI RS'

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1,N_2)	(O_1,O_2)
4	(2,1)	(4,1)
8	(2,2)	(4,4)
	(4,1)	(4,1)

- Active Antenna: CSI RS can be configured as (N_1,N_2) = (2,2)
- The (O_1,O_2) defines the oversampling to be applied in each dimension
- This leads to a grid of 2x4 by 2x4 beams, i.e. a grid of 8x8 beams
- There can be up to 4 CSI RS so up to 64x4 = 256 beams

- RRH + Passive Antenna: each CSI RS is configured as (N_1,N_2) = (4,1)
- This leads to a grid of 4x4 by 1x1 beams, i.e. a grid of 16x1 beams
- Only a single CSI RS is configured so only 16 beams



Number of Beams

- In this case, there are only 8 TRX so there is a one-to-one mapping between TRX and CSI RS Port
- CSI RS cannot be beamformed due to 1 TRX per port



CSI RS Ports

- Vertical Beamforming is not possible so the (4, 1) configuration is selected for (N1, N2)

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1,N_2)	(O_1,O_2)
4	(2,1)	(4,1)
8	(2,2)	(4,4)
	(4,1)	(4,1)

Beam 'Oversampling' in each dimension
i.e. single row of N1 x O1 = 16 beams available for selection during PMI reporting

CSI RS Configuration

- An Active antenna can be configured to broadcast 1, 2 and 4 '8 Port CSI RS'
- A Passive antenna can be configured to broadcast 1 '8 Port CSI RS'
- Number of '8 Port CSI RS' is configured using maxCsirsNumber
- CSI RS for this feature is 'cell specific' so capacity restriction caused by 'UE specific' CSI RS is removed

- The power cannot be set freely and must obey the following rule:

csirsPower = powerControlOffsetSS + *ssPbchBlockPower* + 3 dB

-3, 0, 3 or 6 dB

```
nzp-CSI-RS-ResourceId 0,
resourceMapping
  frequencyDomainAllocation other : '110000'B,
  nrofPorts p8,
  firstOFDMSymbolInTimeDomain 6,
  cdm-Type cdm4-FD2-TD2,
  density one : NULL,
  freqBand
    startingRB 0,
    nrofRBs 108
 ,
powerControlOffset 1,
powerControlOffsetSS db0,
scramblingID 44,
periodicityAndOffset slots80 : 67
```

```
nzp-CSI-RS-ResourceId 182,
resourceMapping
  frequencyDomainAllocation other : '110000'B,
  nrofPorts p8,
  firstOFDMSymbolInTimeDomain 10,
  cdm-Type cdm4-FD2-TD2,
  density one : NULL,
  freqBand
    startingRB 0,
    nrofRBs 108
 ,
powerControlOffset 1,
powerControlOffsetSS db0,
scramblingID 44,
periodicityAndOffset slots80 : 67
```

```
nzp-CSI-RS-ResourceId 181,
resourceMapping
  frequencyDomainAllocation other : '110000'B,
  nrofPorts p8,
  firstOFDMSymbolInTimeDomain 8,
  cdm-Type cdm4-FD2-TD2,
  density one : NULL,
  freqBand
    startingRB 0,
    nrofRBs 108
 ,
powerControlOffset 1,
powerControlOffsetSS db0,
scramblingID 44,
periodicityAndOffset slots80 : 67
,
```

```
nzp-CSI-RS-ResourceId 183,
resourceMapping
  frequencyDomainAllocation other : '110000'B,
  nrofPorts p8,
  firstOFDMSymbolInTimeDomain 12,
  cdm-Type cdm4-FD2-TD2,
  density one : NULL,
  freqBand
    startingRB 0,
    nrofRBs 108
 ,
powerControlOffset 1,
powerControlOffsetSS db0,
scramblingID 44,
periodicityAndOffset slots80 : 67
```

NRCELL - periodicCsiAcquisition

csiReportPeriodicity
csirsNumberWithinReportPeriod_Set1
csirsPower
maxCsirsNumber

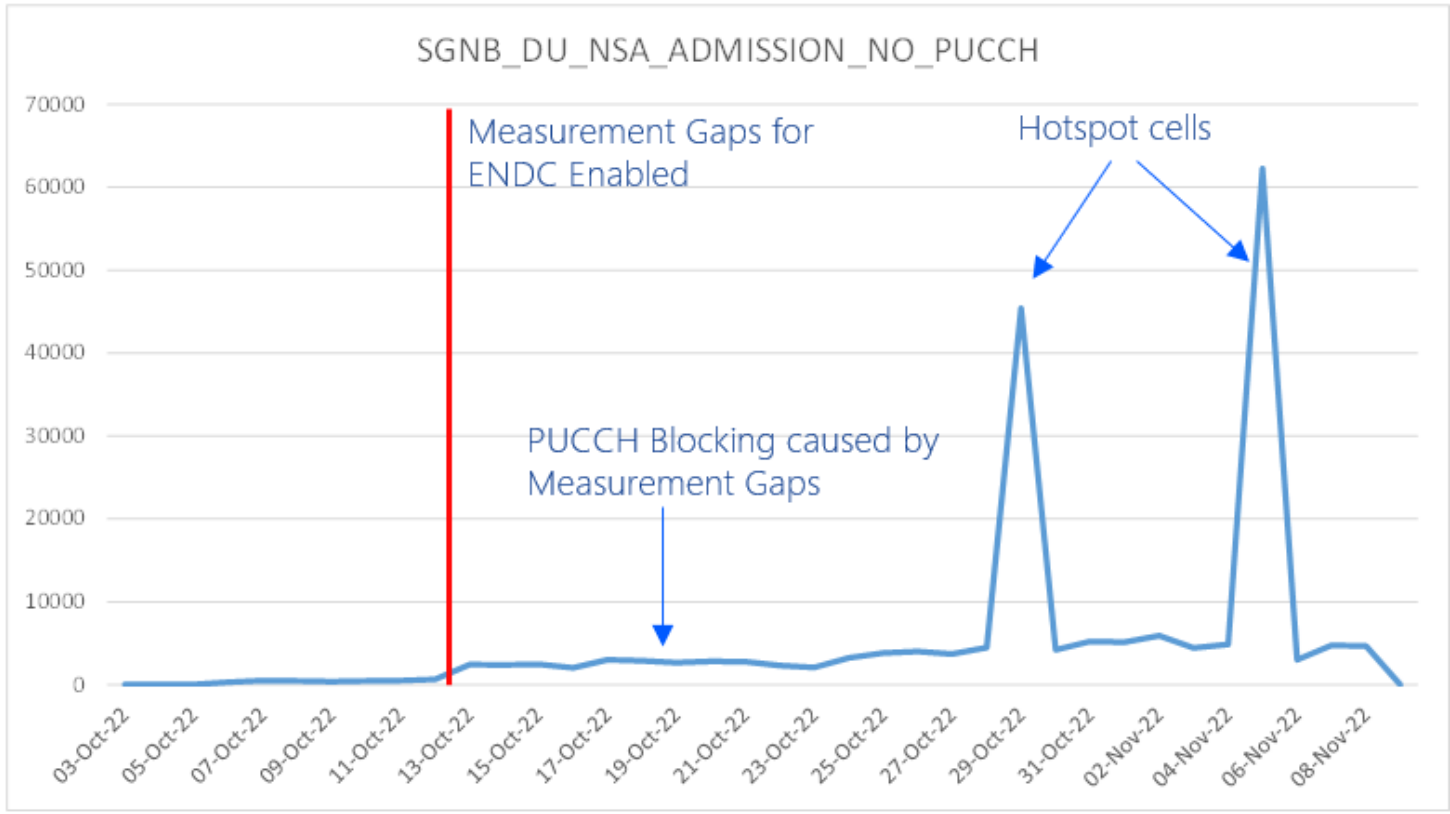
Impact of Measurement Gaps (I)

- Activation of Codebook based Beamforming in combination with Measurement Gaps for ENDC can lead to PUCCH Blocking, even at low levels of traffic
  - 5G Admission Control can can reject new ENDC connection requests if those connections already have Gaps configured at the 4G BTS
  - 5G Admission Control can reject modification requests if Gaps are triggered during an ongoing ENDC connection

- The following 4G parameters are relevant to the activation of Measurement Gaps for ENDC:
  - LNBTS – actMeasGapIfEnDc
  - LNBTS – actMeasGapCoordEnDc
  - LNBTS – actdrxcoordendc
  - LNCEL – actDrxDuringMeasGap

Impact of Measurement Gaps (II)

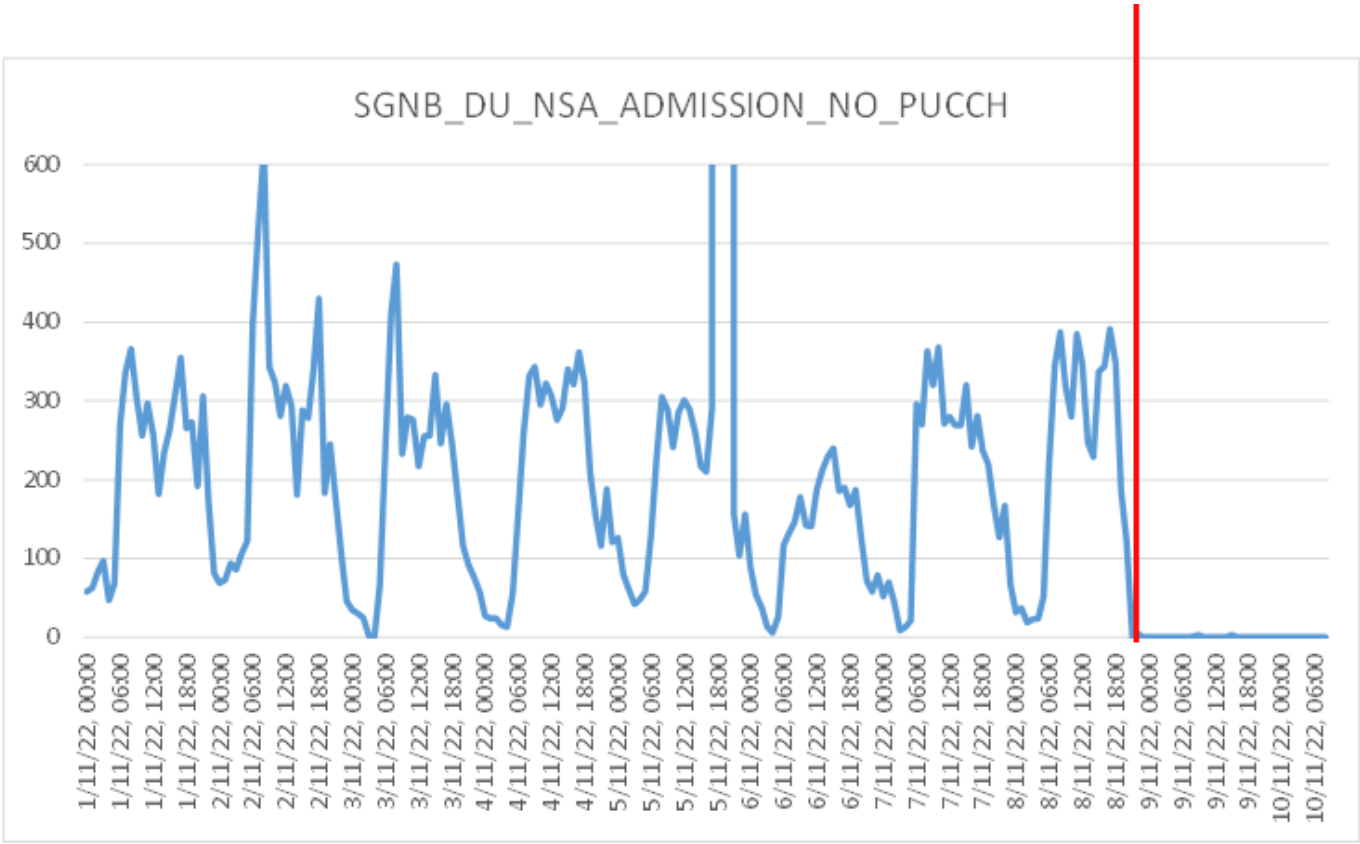
- Some example network statistics illustrating the impact of Measurement Gaps in combination with Codebook based Beamforming are shown to the right
- The higher upwards spikes are not related to the Measurement Gaps. These are generated by high levels of traffic at some individual cells



Impact of Measurement Gaps (III)

- The graph illustrates the impact of increasing ***csirsNumberWithinReportPeriod_Set1*** from '3' to '4'
- This increases the rate of transmission for the CSI RS Set 1 for Codebook based Beamforming so increases the overhead of the CSI RS
- Checks should be completed to determine if there are any negative impacts upon throughputs and spectral efficiency

csirsNumberWithinReportPeriod_Set1 increased from '3' to '4'



SRS based Beamforming

Introduction

- SRS based Beamforming allows the 5G BTS to select appropriate beams based upon measurements from the uplink Sounding Reference Signal (SRS)
- The following features are used to introduce SRS based Beamforming:
  - 5GC001086 SRS based beamforming in UL (22R2)
  - CB008247 SRS-based Downlink Beamforming (22R3)

- SRS based Beamforming continues to use a 'Grid of Beams' (GoB) pattern
  - different Beam Patterns compared to legacy solutions
  - Nokia beam pattern rather than 3GPP beam pattern used by Codebook based Beamforming
  - uses the term '2D DFT' beams

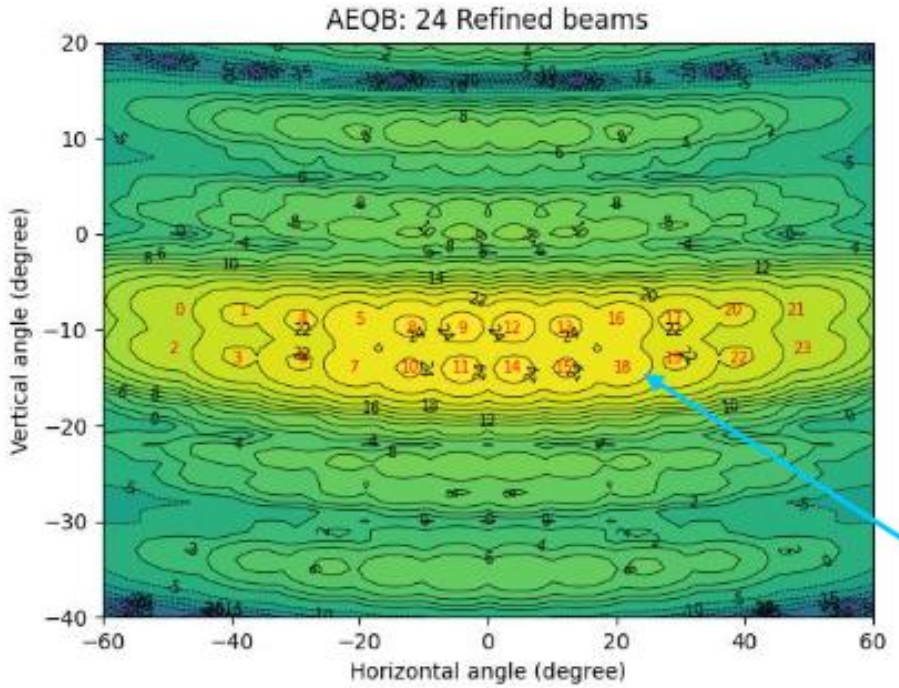
- Supports fallback to legacy solutions if uplink SRS is weak or not received

Requires ABIO Capacity Cards

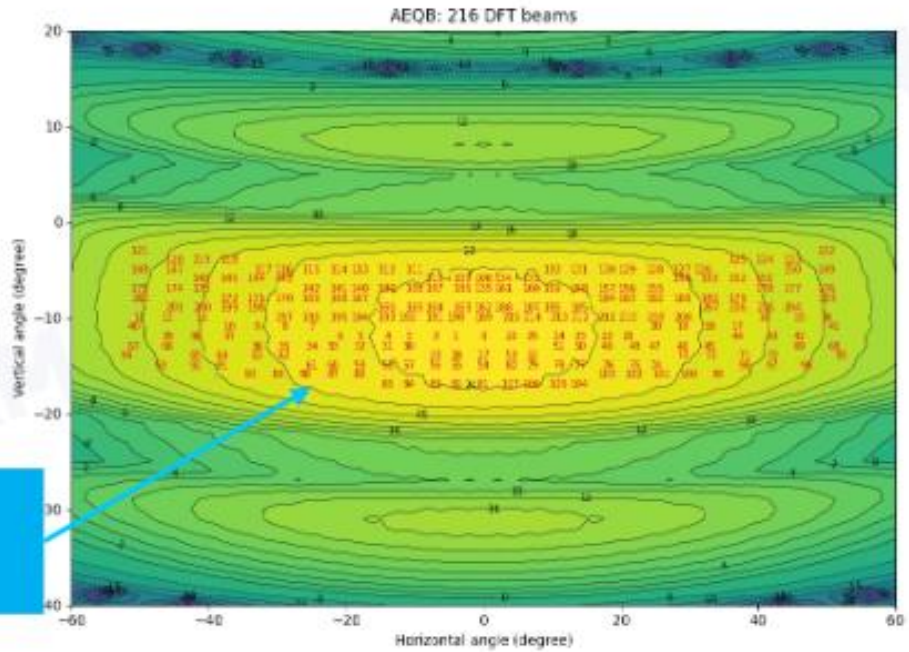
SRS based Beamforming – Grid of Beams (I)

- 2D DFT Grid of Beams provides a significant increase in the number of uplink beams
- Provides an increased area with high gain beams

Legacy Uplink Beamforming



SRS based Uplink Beamforming



Area with +22dBi beam gain

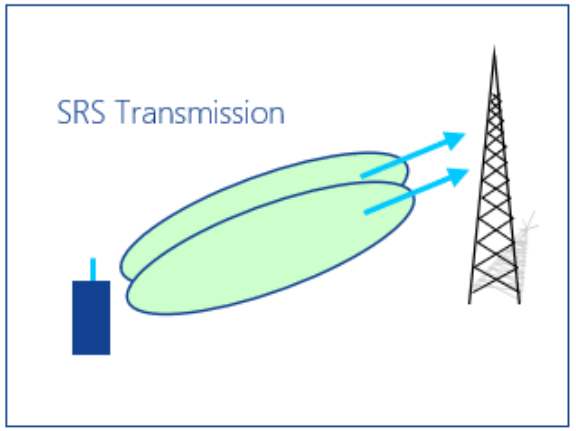
SRS based Beamforming – Grid of Beams (II)

- The number of beams belonging to the beam pattern depends upon both the RF hardware and the parameter configuration
- Some examples are shown below

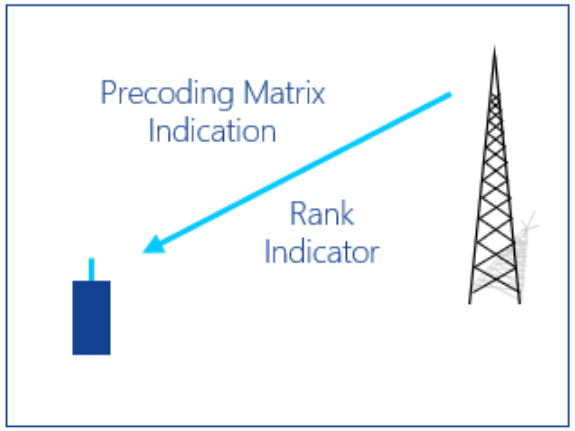
RU type	TRX columns	TRX rows	Horiz. opening angle [deg]	2D DFT columns	2D DFT rows	Number of 2D DFT beams
64TRX AEHC	8	4	90	23	14	322
			120	27	14	378
64TRX AEQK	8	4	90	23	14	322
			120	29	14	406
32TRX AEQZ	8	2	90	21	8	168
			120	27	8	216
32TRX AEQN	8	2	90	23	7	161
			120	29	7	203

SRS based Beamforming in UL – Codebook based Transmission

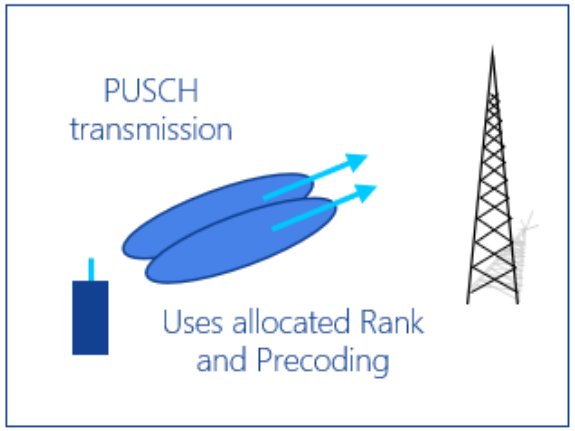
- Does not rely upon channel reciprocity
- Known as 'Codebook' based transmission



- UE transmits SRS
- SRS transmission is not beamformed
- BTS measures SRS to determine the best uplink beams at the BTS and the MIMO Rank/Precoding (if UE supports UL MIMO)



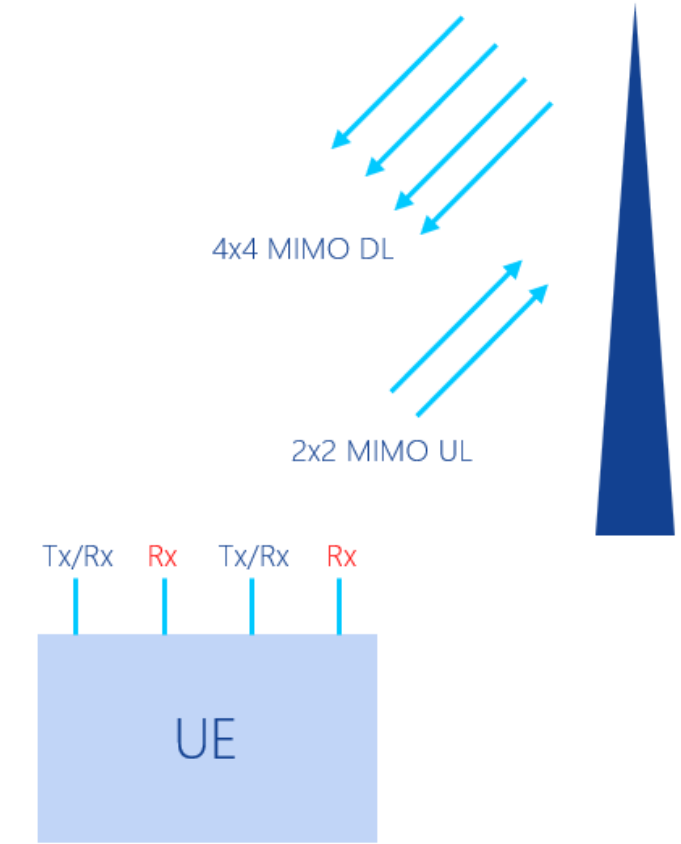
- BTS provides feedback within the PDCCH Downlink Control Information (DCI) used to provide the PUSCH resource allocation



- UE transmits the PUSCH according to the instructions provided by the BTS
- BTS uses receive beams selected from the SRS transmission

SRS based Beamforming in DL - Transmit Antenna Switching (TAS) (I)

- A UE may support:
  - 4x4 MIMO in the downlink: Rx with 4 antenna
  - 2x2 MIMO in the uplink: Tx with 2 antenna
- If the 2 Tx antenna are fixed, the BTS cannot measure the uplink channel from the antenna which are Rx only
- The BTS requires the uplink channel from all UE antenna so it can make proper decisions regarding downlink transmission (assuming channel reciprocity for TDD)
- Transmit Antenna Switching (TAS) allows the UE to change the antenna used for transmission so the BTS can measure the channel from all antenna
- Transmit Antenna Switching is an optional UE capability



SRS based Beamforming in DL - Transmit Antenna Switching (TAS) (II)

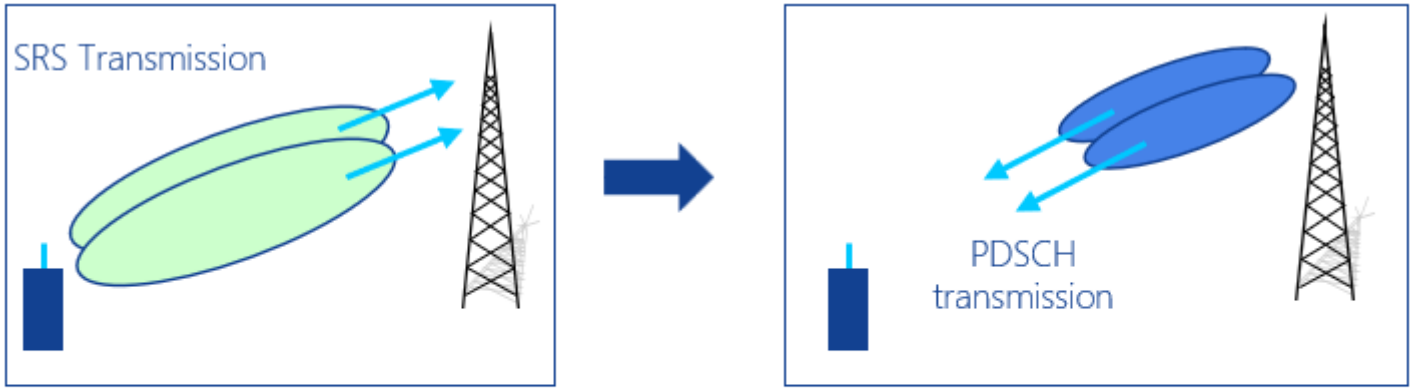
- TAS can cause performance issues when the UE is configured with both FDD and TDD carriers (ENDC with 4G FDD or 5G FDD-TDD CA)
  - UE transmits using an antenna being used for FDD Rx
  - Out-of-Band emissions can generate impact to FDD Rx for specific band combinations
- If TAS is enabled (and supported by UE) then UE has to transmit 2 SRS Resource Sets
  - 1 for Uplink (Codebook SRS Resource Set)
  - 1 for Downlink (Antenna Switching SRS Resource Set)
- Otherwise UE transmits a single Codebook SRS Resource Set for both Uplink and Downlink
- Gain of TAS is currently measured as very small (in the order of 1%)

- Product Line currently recommend to disable TAS using srsTypeAllocForDlBm = Codebook (1)

Parameter	Object	Range	Default
srsForBeamMgmt - srsTypeAllocForDlBm	NRCELL	Antenna Switching (0), Codebook (1), Antenna Switching or Codebook (2)	Antenna Switching or Codebook (2)

SRS based Beamforming in DL – Codebook or Antenna Switching based Transmission

- Relies upon channel reciprocity
- Known as 'Codebook' or 'Antenna Switching' based transmission
- Uplink SRS is only used for Beam Selection
- It cannot be used for Rank nor Precoding selections
  - uplink SRS does not reflect downlink interference conditions
- 2 options are available for Rank, CQI and Precoding Selection (next slide)



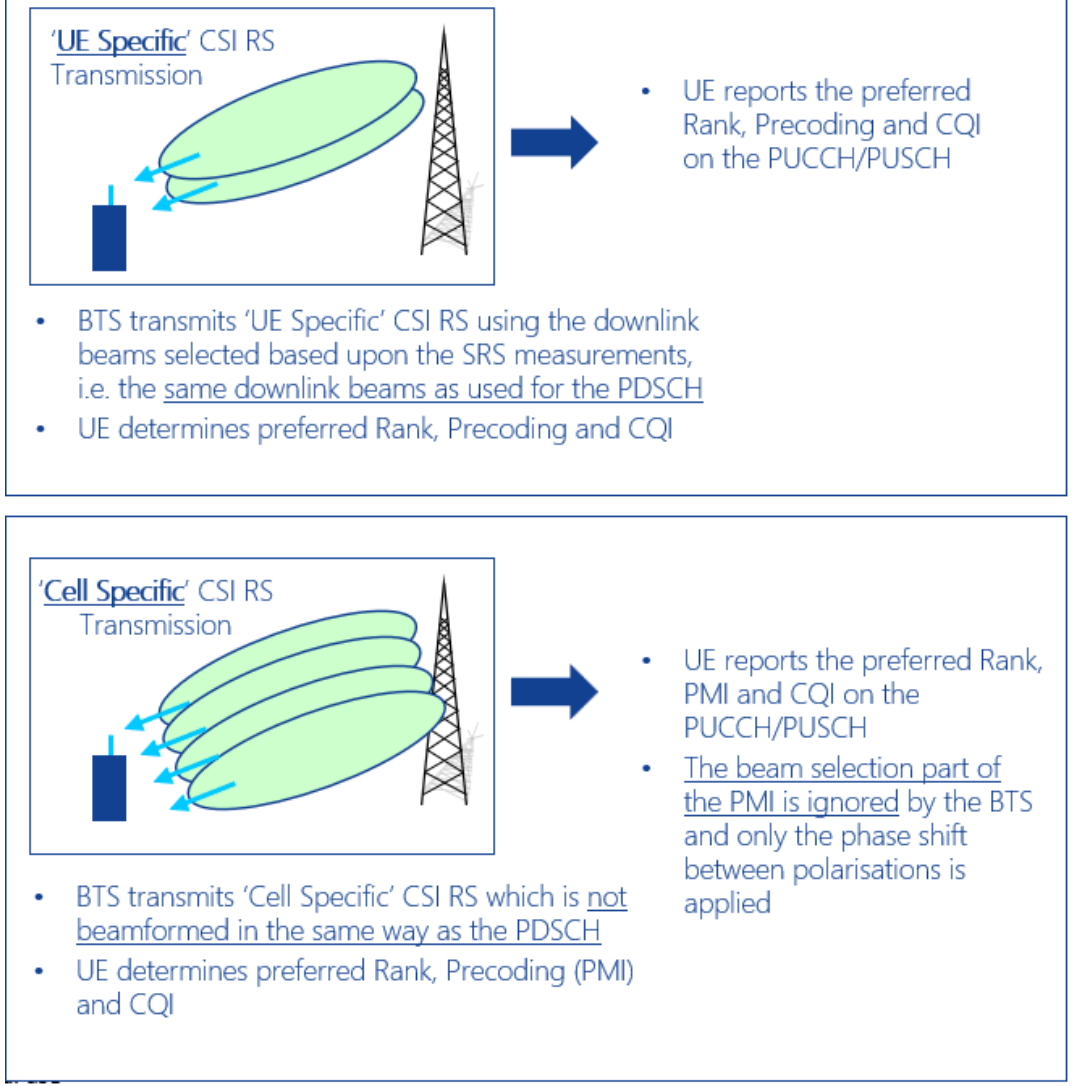
- UE transmits SRS
- SRS transmission is not beamformed
- BTS measures SRS to determine the best downlink beams at the BTS

- BTS transmits the PDSCH using the beams selected from the SRS measurements

SRS based Beamforming in DL – CQI, RI, PMI Reporting

- CSI reporting can be based upon either a 'UE specific' or 'Cell specific' CSI RS
- 'UE specific' solution is more logical because it allows the UE to generate CQI, RI and PMI based upon the same downlink beams as used by the PDSCH
- However, 'UE specific' CSI RS has limited capacity and can lead to connection blocking
- The 'Cell specific' CSI RS solution requires Codebook based Beamforming to be enabled
- Current recommendation from Product Line is to use the 'Cell specific' solution
  - performance impact of not using the PDSCH beamforming for the CSI RS is relatively small
- Can be controlled using the NRCELL **actUeCsirsCoexistCellCsirs** parameter

recommendation to set to 'False'



Frequency Allocation (II)

- C_{srs} specifies row
- B_{srs} specifies column

- Similar approach to 4G
- When using 4G, C_{srs} is cell specific while B_{srs} is UE specific
- When using 5G, both C_{srs} and B_{srs} are UE specific

Wide Bandwidth

Narrow Bandwidth

C_{SRS}	$B_{SRS}=0$		$B_{SRS}=1$		$B_{SRS}=2$		$B_{SRS}=3$	
	$m_{SRS,0}$	N_0	$m_{SRS,1}$	N_1	$m_{SRS,2}$	N_2	$m_{SRS,3}$	N_3
0	4	1	4	1	4	1	4	1
1	8	1	4	2	4	1	4	1
2	12	1	4	3	4	1	4	1
3	16	1	4	4	4	1	4	1
4	16	1	8	2	4	2	4	1
5	20	1	4	5	4	1	4	1
6	24	1	4	6	4	1	4	1
7	24	1	12	2	4	3	4	1
...	...	...	...	...	...	...	...	...
30	128	1	64	2	32	2	4	8
31	128	1	64	2	16	4	4	4
32	128	1	16	8	8	2	4	2
33	132	1	44	3	4	11	4	1
34	136	1	68	2	4	17	4	1
...	...	...	...	...	...	...	...	...
56	240	1	24	10	12	2	4	3
57	256	1	128	2	64	2	4	16
58	256	1	128	2	32	4	4	8
59	256	1	16	16	8	2	4	2
60	264	1	132	2	44	3	4	11
61	272	1	136	2	68	2	4	17
62	272	1	68	4	4	17	4	1
63	272	1	16	17	8	2	4	2

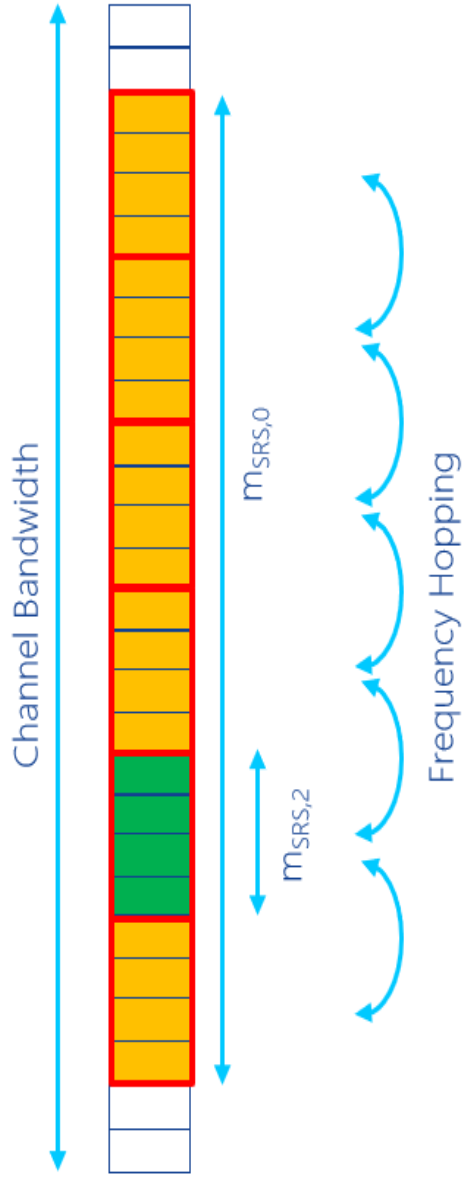
Number of Resource Blocks

Resource Block Multiple for column towards the left

Frequency Allocation (III)

- $m_{SRS,0}$ specifies the total bandwidth of the SRS after frequency hopping
  - value is selected by configuring *Csrs*
  - should be selected to be ≤ PUSCH bandwidth after accounting for Long PUCCH overhead
- $m_{SRS,x}$ specifies the bandwidth of a single specific SRS transmission
  - value selected by configuring *Bsrs*

C_{SRS}	$B_{SRS}=0$		$B_{SRS}=1$		$B_{SRS}=2$		$B_{SRS}=3$	
	$m_{SRS,0}$	N_0	$m_{SRS,1}$	N_1	$m_{SRS,2}$	N_2	$m_{SRS,3}$	N_3
0	4	1	4	1	4	1	4	1
1	8	1	4	2	4	1	4	1
2	12	1	4	3	4	1	4	1
3	16	1	4	4	4	1	4	1
4	16	1	8	2	4	2	4	1
5	20	1	4	5	4	1	4	1
6	24	1	4	6	4	1	4	1
7	24	1	12	2	4	3	4	1



SRS based Beamforming – Connection Capacity

- The SRS for Beam Management is only configured within 1 symbol of the Special Slots
  - prevents the SRS from impacting application throughput because Special Slots are not currently used for uplink data transfer
- The capacity depends upon the channel bandwidth, the SRS bandwidth and the use of TAS
- If TAS is disabled then each UE requires only 1 SRS Resource Set
- If the Channel Bandwidth is 40 MHz (106 PRB), the total SRS hopping bandwidth is 96 PRB and the SRS bandwidth per transmission is 16 PRB then 6 UE can be frequency multiplexed at PRB level
- Nokia implementation uses a subcarrier Transmission Comb of 2 and 4 Cyclic Shifts, so 2 x 4 x 6 = 48 UE can be frequency/code multiplexed per slot
- With an SRS period of 40 slots, and 4 Special slots per 40 slots then the capacity is 192 connections

Periodic SRS is only allocated in special slots																				
Subframe	0		1		2		3		4		5		6		7		8		9	
Slot	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1208	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U	D	D	D	S	U
1070	D	D	D	S	U	D	D	S	U	U	D	D	D	S	U	D	D	S	U	U
1116_0ms	D	D	D	S	U	U	D	D	D	D	D	D	D	S	U	U	D	D	D	D
1116_3ms	D	D	D	D	D	D	D	S	U	U	D	D	D	D	D	D	D	S	U	U

Recommended SRS Configurations

1st candidate								2nd candidate					
Channel Bandwidth (chBw)	PRB amount	SRS Bandwidth (cSRS)	SRS Subband (bSRS)	SRS frequency range	Subband size	SRS Capacity per slot	SRS vs BW	SRS Bandwidth (cSRS)	SRS Subband (bSRS)	SRS frequency range	Subband size	SRS Capacity per slot	SRS vs BW
20	51	**12**	**2**	48	**8**	48	94.12%						
30	78	**19**	**2**	72	**12**	48	92.31%	**20**	**2**	76	**4**	152	97.44%
40	106	**23**	**2**	96	**16**	48	90.57%	**25**	**2**	104	**4**	208	98.11%
50	133	**32**	**2**	128	**8**	128	96.24%						
60	162	**38**	**2**	144	**8**	144	88.89%	**42**	**2**	160	**16**	80	98.77%
70	189	**42**	**2**	160	**16**	80	84.66%	**45**	**2**	184	**4**	368	97.35%
80	217	**49**	**2**	192	**8**	192	88.48%						
90	245	**55**	**3**	240	**8**	240	97.96%						
100	273	**63**	**2**	272	**8**	272	99.63%						

- The defined values represent balance between the SRS capacity and performance.
- If no specific requirement e.g. single UE peak throughput test, then 1st candidate should be used as default settings.

NOTE:

- SRS subband size **8-16 PRB** is **the first recommendation**
- **4 PRB works fine in simulation** and is available for field trials.
- **Smaller SRS** subband size provides **higher SRS capacity** and **wider coverage**. 4PRB can be used if it works fine in field after testing

Fallback to Legacy Beamforming

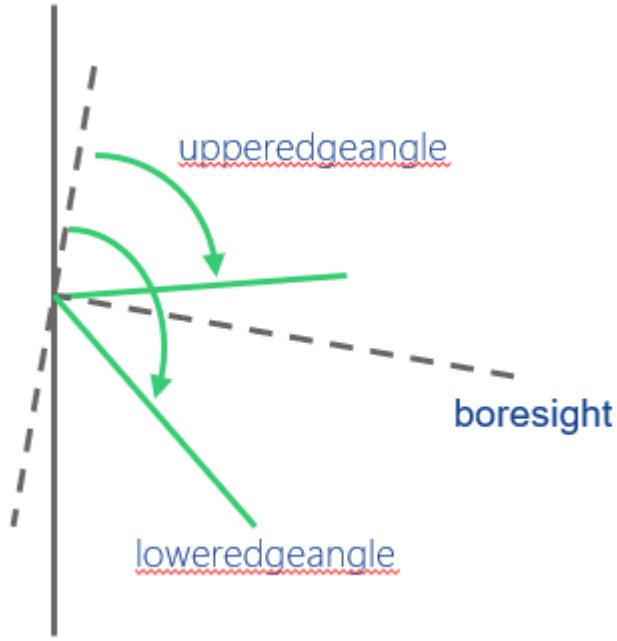
- SRS based Beamforming may not perform well towards the cell edge due to the limited coverage of the uplink SRS
- The srsPowerThresholdForDlBm parameter allows an RSRP threshold to be configured for fallback to a legacy downlink beamforming solution (Codebook based Beamforming or SS/PBCH)

Parameter	Object	Range	Default
srsForBeamMgmt - srsPowerThresholdForDlBm	NRCELL	-150 to 0 dBm	-120 dBm

- This parameter is introduced with 23R1
- 22R3 and 22R4 use NRCELL - tmpParam3 to provide the same functionality
- The NRCELL actUeCsirsCoexistCellCsirs parameter impacts the fallback solution
  - 'False' leads to fallback to Codebook based Beamforming
  - 'True' leads to fallback to SS/PBCH based Beamforming

5GC001086 SRS based Beamforming in UL – Upper/Lower Edge Angle

- 5GC001086 introduces support for upper and lower edge angles for FR1 TDD
  - previously, the upper and lower edge parameters were only used for Analogue Beamforming in FR2
- Recommended values are:
  - upperedgeangle = 80 degrees
  - loweredgeangle = 135 degrees



Parameter	Object	Range	Default
beamset - loweredgeangle	NRCELL	95 to 135 degrees (digital beamforming) 45 to 135 degrees (analogue beamforming)	120 degrees
beamset - upperedgeangle	NRCELL	45 to 90 degrees (digital beamforming) 45 to 135 degrees (analogue beamforming)	60 degrees

If **NRCELL.beamSet.basicBeamSet** is equal to 9 (corresponding to beamSet_2_2_2_2),
then these parameters can be equal to 20 or 45 (respectively for 40° and 90° cells)
else these parameters can be equal to 45 or 60 (respectively for 90° and 120° cells).
In addition, before more sophisticated beamforming settings are available,
NRCELL.beamSet.leftEdgeAngle and **NRCELL.beamSet.rightEdgeAngle must have the same value**.

5GC001086 SRS based Beamforming in UL – Antenna Array

- The *mMIMOAntArrayMode* parameter determines the allocation of Active Antenna TRX to a specific cell
- The 'Left/Right' split is not permitted when using SRS based Beamforming in the UL
- Either the full antenna array must be allocated or a 'Top/Bottom' split

Parameter	Object	Range	Default
mMIMOAntArrayMode	NRCELL	Full 64TRX Array (4x8x2) (0), ~~Left 32TRX Array (4x4x2) (1), Right 32TRX Array (4x4x2) (2),~~ Top 32TRX Array (2x8x2) (3), Bottom 32TRX Array (2x8x2) (4), Top 16TRX Array (1x8x2) (5), Bottom 16TRX Array (1x8x2) (6), Full 32TRX Array (2x8x2) (7)	-

CB10602 SRS based Beamforming Extension (I)

- 22R4 supports downlink power balancing using CNI-84878
  - enabled using bit 16 of NRCELL - *tmpActFeat1*
  - allows PDSCH power balancing for Rank 4 when using the Cell specific CSI-RS
- 23R1 supports downlink power balancing using CB10602
  - allows PDSCH power balancing for both Ranks 3 and 4
    - applicable when using either the Cell specific or UE specific CSI RS
  - also allows power balancing for the UE specific CSI RS

Parameter	Object	Range	Default
actCsirsBeamPowerBalance	NRCELL	False (0), True (1)	False (0)
actpdschbeampowerbalance	NRCELL	False (0), True (1)	False (0)
beampoweroffsetthreshold	NRCELL	0 to 15, step 1 dB	6 dB

Power balancing for the UE Specific CSI RS

Power balancing for the PDSCH

Limits the upper level of Power Balancing

CB10602 SRS based Beamforming Extension (II)

- The 5G BTS measures the imbalance between beams using the uplink SRS
- A row from one of the tables below is selected based upon the measured imbalance
  - the row which has the closest larger value of *bestBeamPowerOffset* is selected
  - for example, if the measured imbalance is -1.5 dB then *bestBeamPowerOffset* = -1 dB is selected

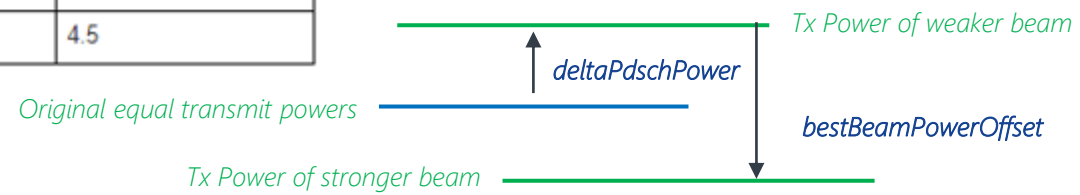
Rank 3

index	bestBeamPowerOffset [dB]	deltaPdschPower [dB]
0	0	0
1	-1	0.4
2	-2	0.8
3	-3	1.2
4	-4	1.5
5	-5	1.8
6	-6	2
7	-8	2.3
8	-10	2.5
9	-15	2.8

Rank 4

index	bestBeamPowerOffset [dB]	deltaPdschPower [dB]
0	0	0
1	-1	0.6
2	-2	1.2
3	-3	1.7
4	-4	2.2
5	-5	2.6
6	-6	3.0
7	-8	3.5
8	-10	3.9
9	-15	4.5

- If the power imbalance is larger than the value of NRCELL – *beampoweroffsetthreshold* then it is rounded down to the value of *beampoweroffsetthreshold*
- The power of the weaker beam is increased by *deltaPdschPower* while the power of the stronger beam is *bestBeamPowerOffset* below the power of the weaker beam



Sounding-based DL Short-term Sub-band BF for NR FR1 TDD

DL Performance

Future

Advanced real-time sub-band beamforming for SU-MIMO

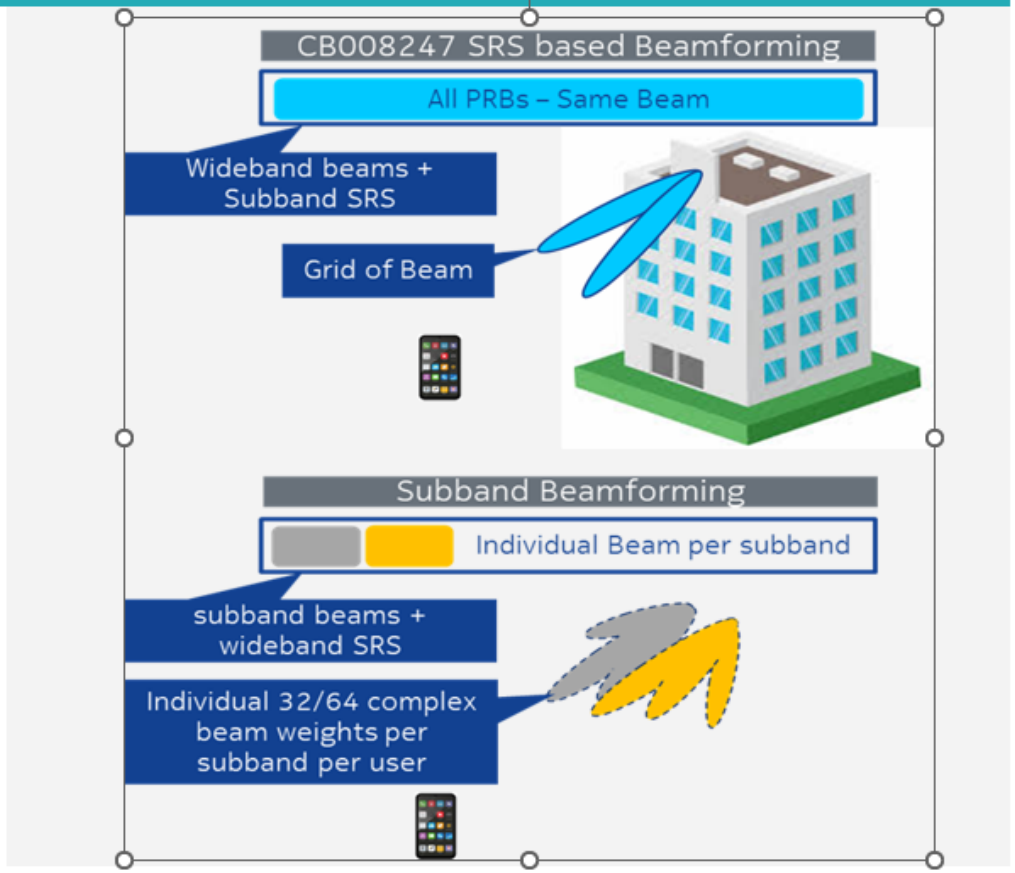
Feature highlights:

- New sounding-based beamforming for SU-MIMO with wide-band SRS, (max 4/CC) qualified users are selected: 1) with TAS capability; 2) Good channel conditions; 3) enough data in the buffer; 4) low mobility / low channel change
- Real-time beam weights vectors downloaded to RU dynamically
- Osprey delivery in 24R1, max 2CC if 8 FH streams configured; max 4CC if 4 FH streams configured; as well as mixed cases but no 16 FH streams support for all CCs of the radio with at least one CC CB008322 enabled
- All CA cases supported but O&M can disable new algorithm for FDD-TDD UL CA, TDD-FDD DL CA & TDD-FDD UL
- New counters added to reflect the new algorithm scheduling

Benefit for customers:

- Increased downlink UE peak throughput

CB008322



SRS-based Downlink Beamforming for Intra-band TDD CA

DL Performance

Future

Enable SRS based BF in SCell for intra-band TDD CA

Enable the SRS assisted BF for CA user in Scell.

Support both SU MIMO and MU MIMO for SCell based on SRS channel estimations from PCell for CA users.

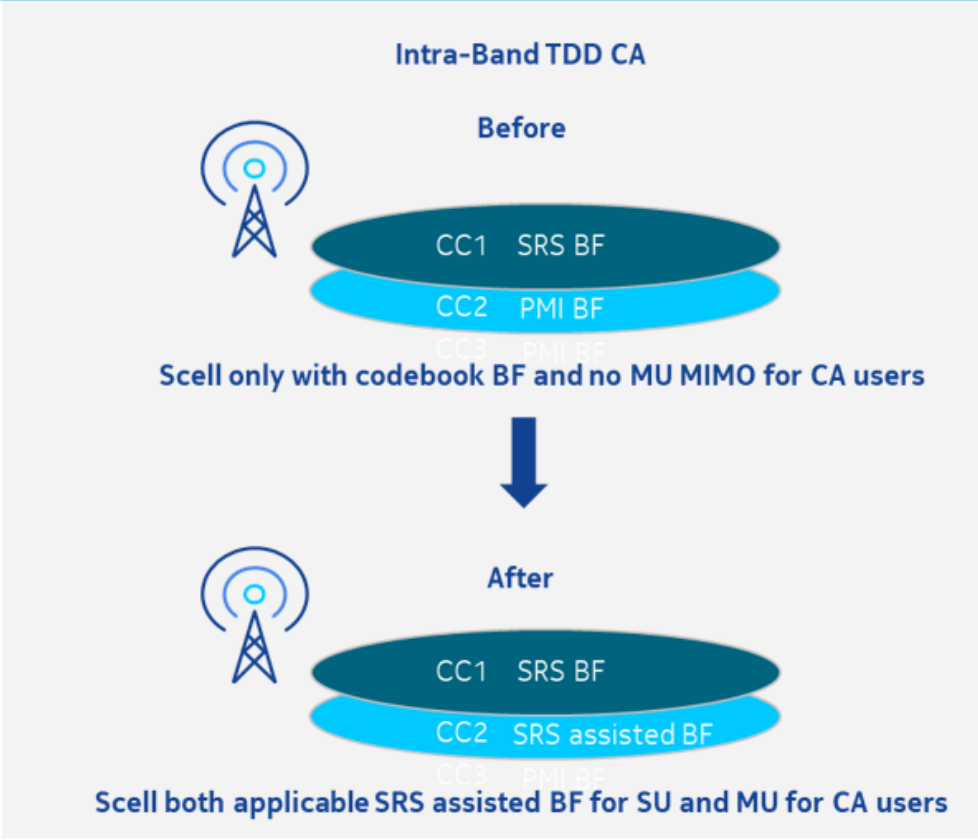
Use case co-coverage between CCs and intra-band CA.

Platform: TDD FR1 ABIO/N and mMIMO 32/64 TRX radios.

Benefits:

- Enable SRS assisted BF for SCell and enhance the spectrum efficiency for CA users

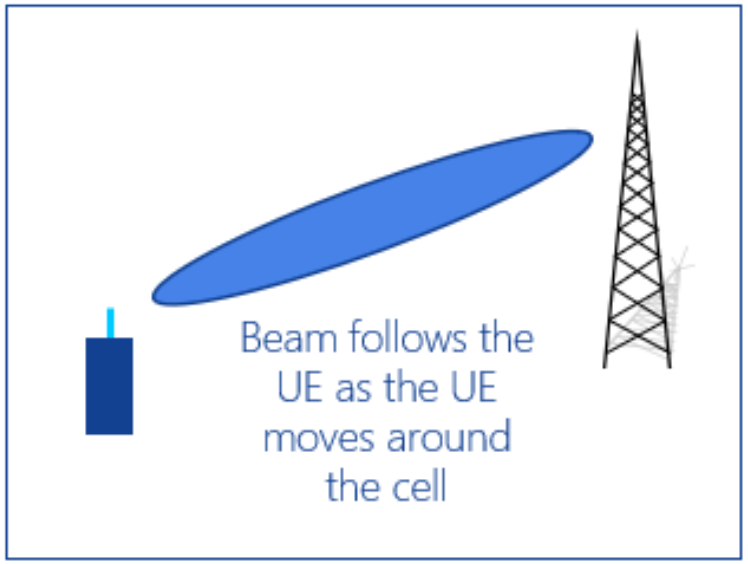
CB008898



Eigenmode Beamforming

Eigenmode Beamforming

- Eigenmode Beamforming does not rely upon a fixed ‘Grid of Beams’
- The 5G BTS dynamically calculates the optimal beam position as the UE moves around the cell, i.e. the beam follows the UE
  - computationally challenging for a large number of UE
- Not currently visible on Nokia roadmap



Unlocking the Full Potential of Mobile

